AUE 8550 VEHICLE STRUCTURAL ANALYSIS FOR AUTOMOTIVE SYSTEMS

Prof. David William Schmueser

TERM COURSE PROJECT

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Team_07

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BIW STRUCTURE DEVELOPMENT FOR NEW VEHICLE DESIGN

Part 1: BIW Structural Lay-Out

The objective of this project was to construct a structural beam frame that satisfied passenger capacity requirements while providing enough packing space for the vehicle's components. This had to be achieved while maintaining a lightweight and satisfying the safety standards for bending, torsion, and crashworthiness.

For this section, it is assumed that each component of the frame is a straight beam with a square or rectangular cross-section.

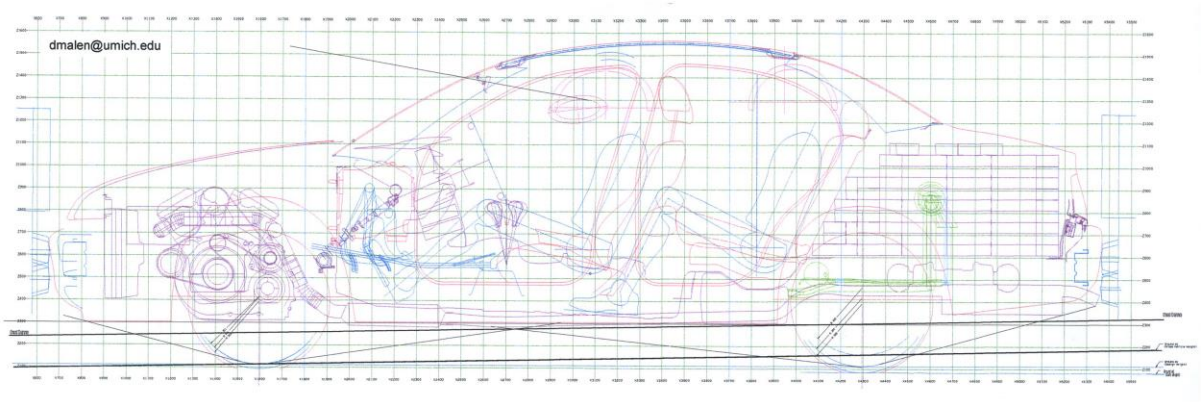


Figure 1 Side View

We can determine the following data from the preliminary vehicle layout:

- Total length (bumper to bumper) = 4720mm
- Wheelbase = 2700mm
- Front track width = 1550mm
- Rear track width = 1550 mm

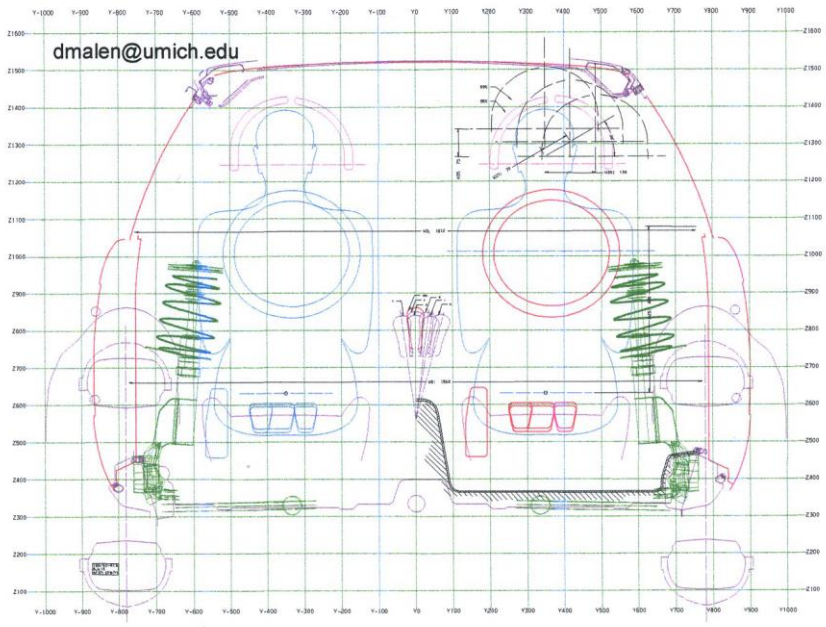


Figure 2 Front View

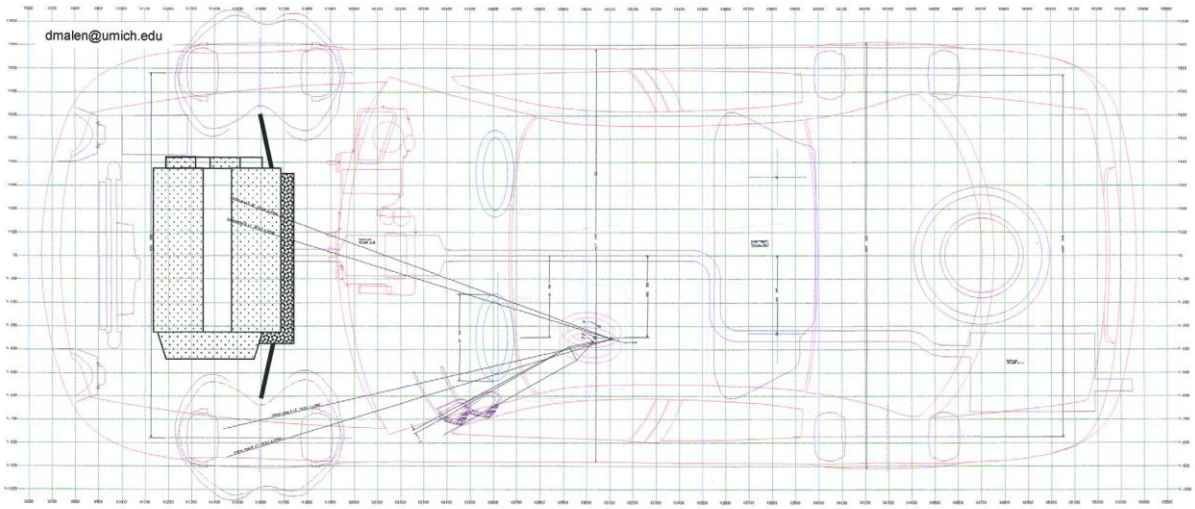


Figure 3 Top View

To determine the baseline values for the side frame profiles, the side view and front view data of the car are used as a guide:

- The floor-to-head distance = 1135mm
- The front mid-rail = 1050mm

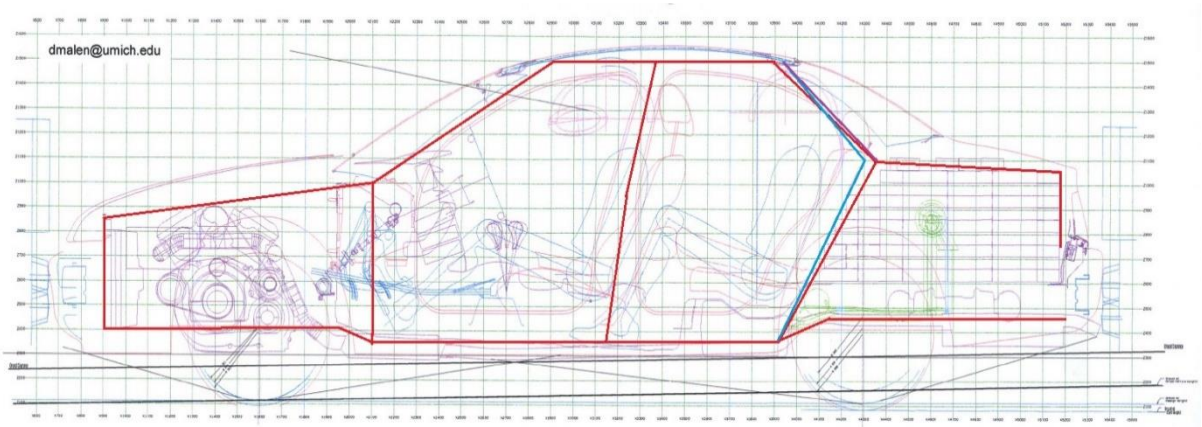


Figure 4 Layout

We have drawn the lines on the side frames of the structure in order to determine the length of each pillar using the assumption that one unit is 10mm X 10mm.

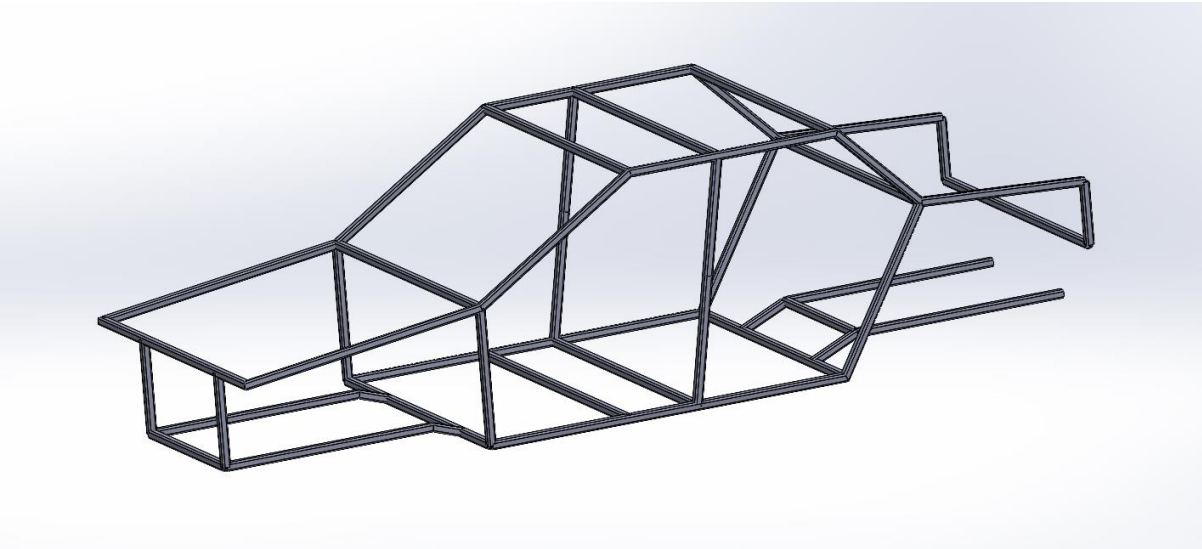


Figure 5 Isometric View of BIW Structure

S.No	Beam	Length
1	Hinge Pillar	650mm
2	A-pillar	945mm
3	Roof rail	1050mm
4	C-pillar	565mm
5	Dog leg pillar	875mm
6	B-pillar upper	560mm
7	B-pillar lower	608mm
8	Rocker	1825mm
9	Front upper rail	1210mm
10	Mid rail	1050mm
11	Upper radiator support	1550mm
12	Lower radiator support	800mm
13	Vertical radiator beam	450mm
14	Front kick-down rail	160mm
15	Rear kick-down rail	250mm
16	Rear Mid rail	1050mm
17	Rear Up rail	832mm
18	Cross member beams (1-5)	1550mm

Table 1 Dimensions of the Beams

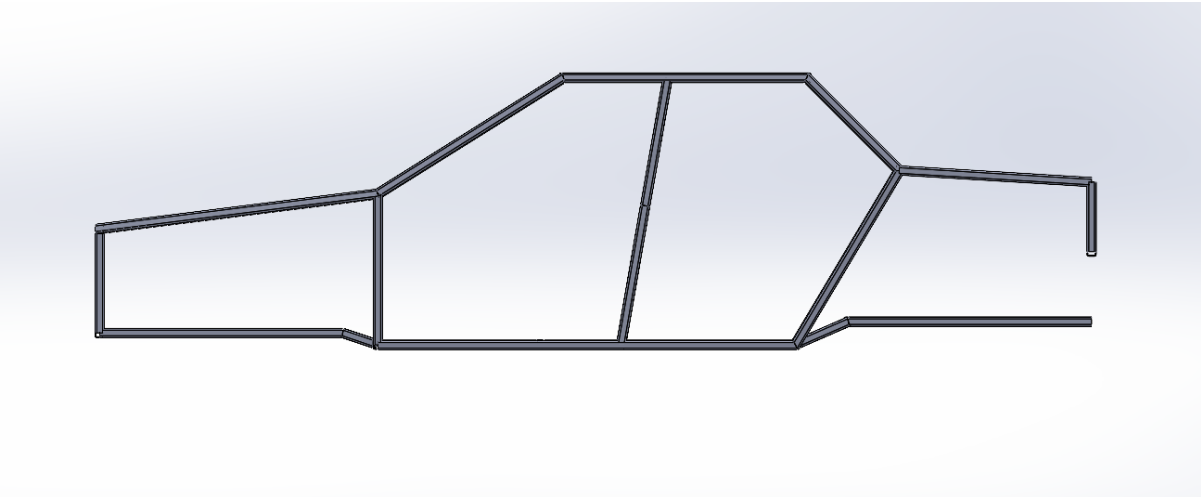


Figure 6 Side View of BIW Structure

Part 2: Body Side-Frame Design for Bending Stiffness

Assignment 1

Taking into account the geometric information from the B-Pillar-to-Rocker joint design for the 2016 Honda Civic. The thickness, t=1.0 mm, of the stamped B-Pillar-Rocker joint. A linear, 2D shell-based HyperWorks finite element model was created using this joint geometry to calculate the rotational joint stiffness for the B-Pillar-to-Rocker Joint. The following part will employ Joint G in conjunction with this rotational joint stiffness. The following qualities of steel, which is considered to be a joint material, are included:

- E = 210 GPa
- N (Poisson's Ratio) = 0.3
- P (Density) = 7890 kg/m3

Quad elements were used to form the mesh, which was then cleaned up to fix the problematic elements. The edges of the rocker were secured, and a load of 1000 N was applied to the top edge of the b-pillar. The meshing's quality metrics were as follows:

Quality metrics:

- Max. Warpage = 14.9
- Max. Aspect Ratio = 4.4
- Max. Skew = 39.9
- Min. Jacobian = 0.62
- Min. Interior Quad Angle = 42.5
- Max. Interior Quad Angle = 132.1
- Min. Interior Tri Angle = 30
- Max. Interior Tri Angle = 105.3

Range

- Warpage >15
- Aspect Ratio >5
- Skew >40
- Jacobian <0.6
- Min. Interior Quad Angle <40
- Max. Interior Quad Angle >140
- Min. Interior Tri Angle <30
- Max. Interior Tri Angle >120

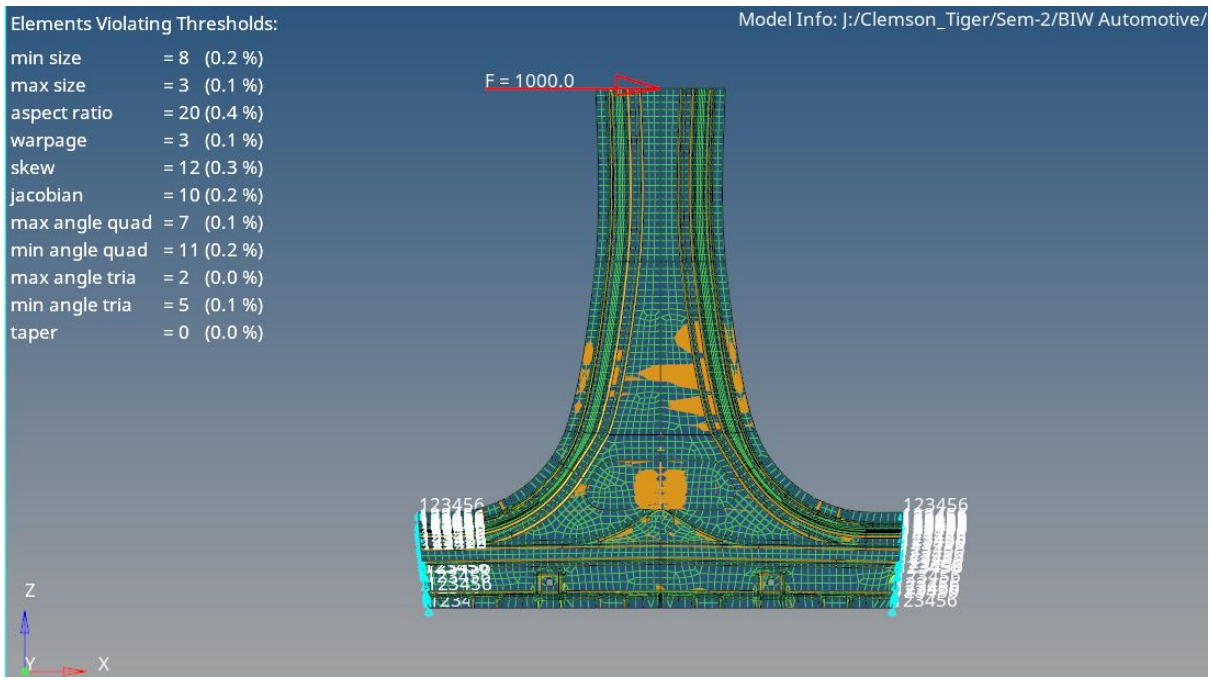


Figure 7 Meshing and Boundary Conditions

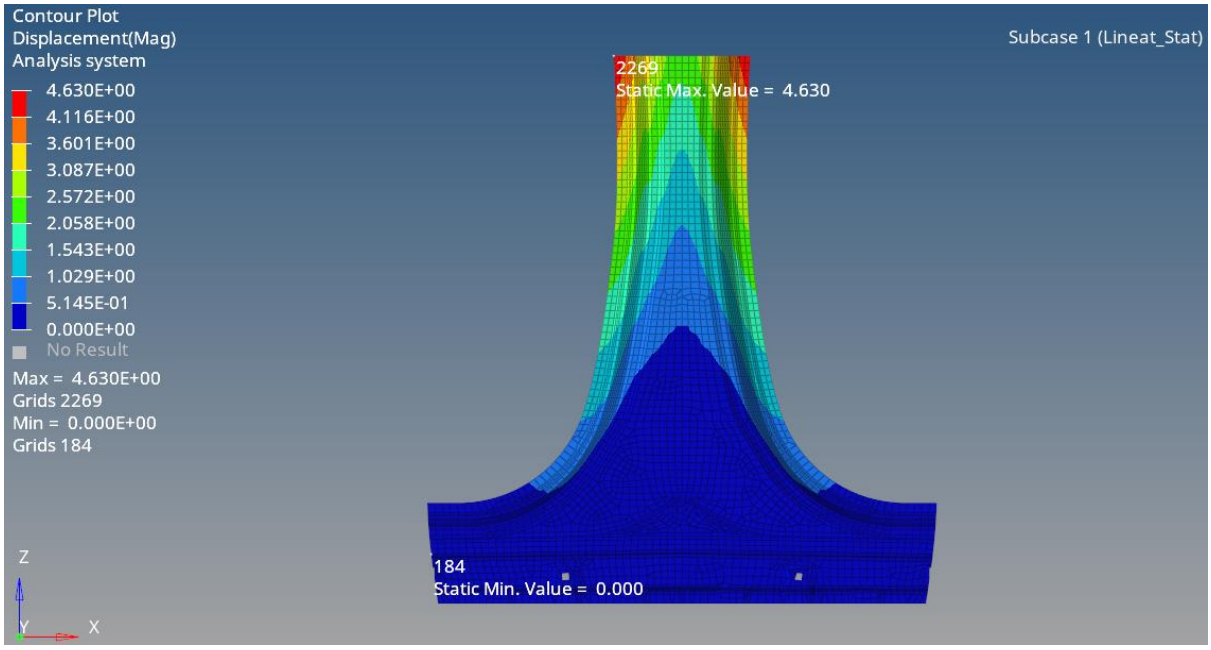


Figure 8 Contour plot of the displacement with Min and Max displacement value

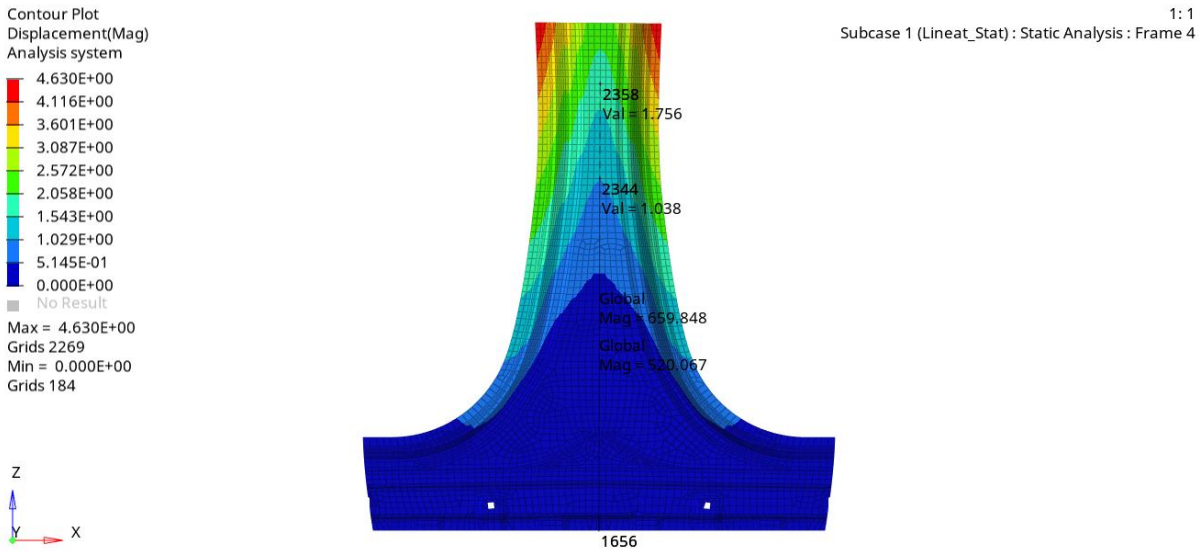
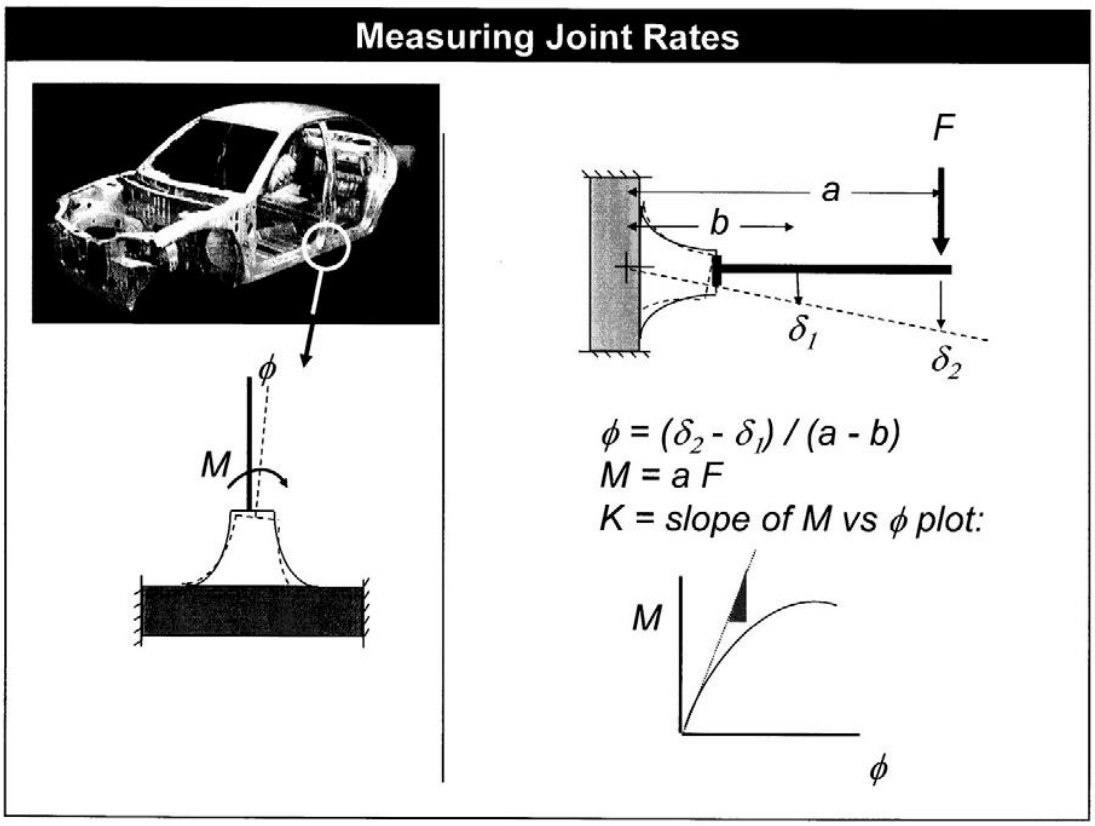


Figure 9 Location of nodes and distance

Following drafting the displacement contour, we chose two nodes and measured the magnitude of the displacement at those locations and the distance between them. This is the method we use to determine the stiffness of the joint G during post-processing.



(c) Donald E. Malen, dmalen@umich.edu

Figure 10 Measuring Joint rates

The explanation shows that the value is obtained by dividing the deflection on the two nodes by their separation.

In our scenario,

- $\delta 1$=1.038mm ; $\delta 2$=1.756mm
- a=659.848mm; b=520.067mm

$$\phi = \frac{\delta_2 - \delta_1}{a-b}$$

$$\phi = \frac{1.756-1.038}{659.848-520.067}$$

$$\phi = 5.136 \times 10^{-3}$$

$$M = a \times F$$

$$M = 659.848 \times 1000$$

$$M = 659848 \text{ Nm} = 6.6 \times 10^{5} \text{ Nmm}$$

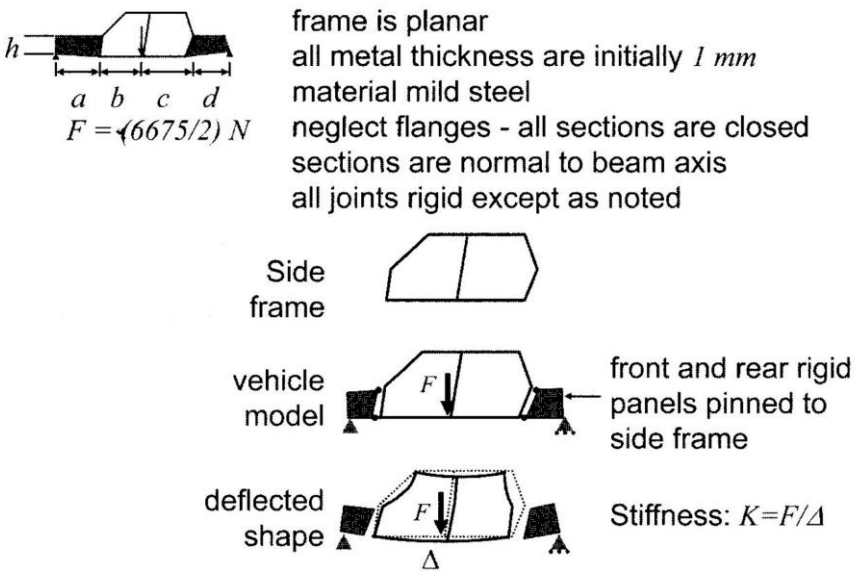
$$K = \frac{M}{\phi}$$

$$K = \frac{6.6*10^{5}}{5.136*10^{-3}}$$

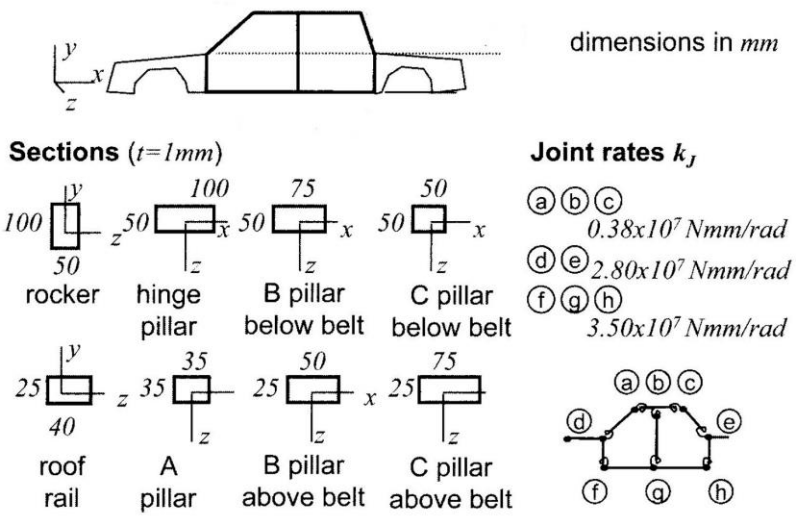
$$K = 1.285 \times 10^{8} \text{ Nmm/rad}$$

Assignment 2

We now estimate the body side-structure bending stiffness utilizing the side frame of the body from the previous section and the stiffness of the joint G discovered by post-processing. The benchmarking results indicated that 7,000 N/mm of stiffness is needed for the vehicle.



We may expand the size of the beams by 200% to achieve the bending stiffness requirement (with the exception of the rocker section, which is limited to 125% for both entry and exit). Each section's thickness is adjustable up to 3 mm.



Input Data

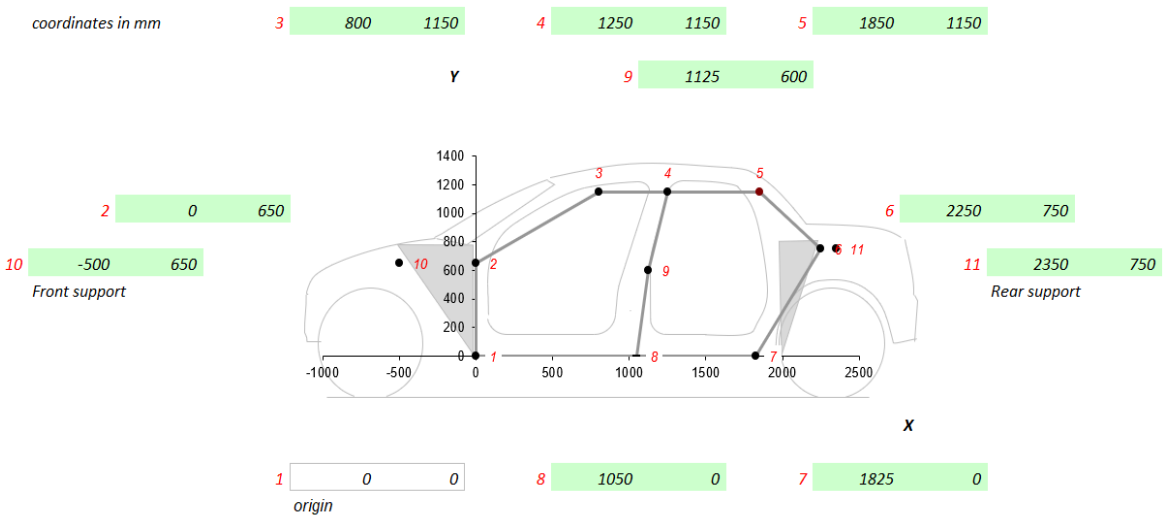


Figure 11 Input coordinates of the side frame geometry

Material	E	G	ρ
	N/mm²	N/mm²	kg/mm³
	207000	78000	7.830E-06

Roof Rail h=	25	Izz= 15104.2	mm⁴	
w=	40	A= 130	mm²	
t=	1.0	mass= 1.27	kg	

B-Upper h=	50	Izz= 52083.3	
w=	25	A= 150	
t=	1.0	mass= 0.59	

Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad

A Pillar h=	35	Izz= 28583.3	
w=	35	A= 140	
t=	1.0	mass= 0.78	

C-Upper h=	75	Izz= 140625.0	
w=	25	A= 200	
t=	1.0	mass= 0.88	

Kj=	2.80E+07	Nmm/rad
Kj=	2.80E+07	Nmm/rad

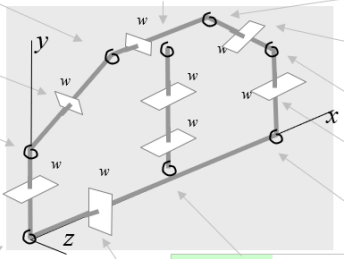
Hinge Pillar h=	100	Izz= 416666.7	
w=	50	A= 300	
t=	1.0	mass= 1.76	

C-Lower h=	50	Izz= 83333.3	
w=	50	A= 200	
t=	1.0	mass= 1.17	

Kj=	3.50E+07	Nmm/rad
Kj=	1.29E+08	Nmm/rad
Kj=	3.50E+07	Nmm/rad

Rocker h=	100	Izz= 416666.7	
w=	50	A= 300	
t=	1.0	mass= 4.70	

B-Lower h=	75	Izz= 210937.5	
w=	50	A= 250	
t=	1.0	mass= 1.47	



w is width into screen (mm)
h is section height (mm)
t is thickness (mm)

Side frame beam mass= 12.61 kg
(one side)

Figure 12 Initial values of pillars.

Side Frame Restrains and Loads

Node Restraints			
Node	X	Y	Rotation
10	F	F	
11		F	

Add node number and F to fix the deflection

Loads (N)			
Beam	Position	Direction	Magnitude
9	0	Y	-3337.5

Add beam number, position of load, direction (X or Y)
Position=0 at beginning of beam (base of arrow on diagram)
Add magnitude (may be negative)

For bending test use

10	F	F
11		F

For effective shear rigidity use

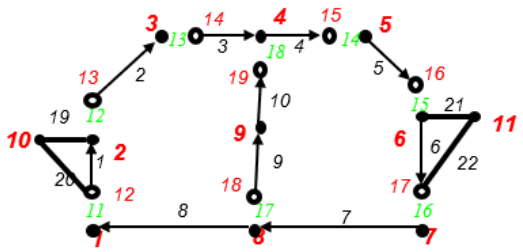
1	F	F
7		F

For bending test use

9	0	Y	magnitude

For effective shear rigidity use

4	0	X	magnitude



numbers in black=beams
numbers in bold red=primary nodes
numbers in green=joints
numbers in small red=secondary nodes

Apply loads to beams
Apply restraints to nodes

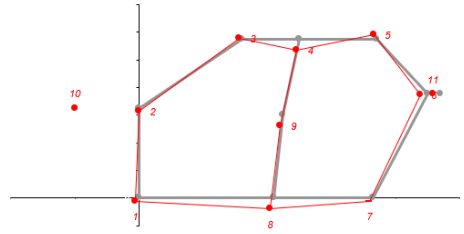
Used for joint stiffness connections

Figure 13 The restraints, loads - direction and magnitude

The nodes 10 and 11 were constrained, and a load of 6675N was divided between them, with 3337.5N on each side in the downward y direction.

Back to Geometry

Side Frame Results-Deflections, Energy, Stress



Deflection scale 20

	Deflections (mm) x	Deflections (mm) y
1	-1.45	-1.10
2	0.00	-1.11
3	-0.97	0.41
4	-0.98	-4.07
5	-1.01	1.73
6	-2.96	-0.22
7	-1.41	-1.06
8	-1.43	-4.02
9	-1.32	-4.02
10	0.00	0.00
11	-2.96	0.00

	Bending Energy Nmm	bending energy %	Bending Stress Nmm² end 1	Bending Stress Nmm² end 2	total stress Nmm²	Cap buckling stress Nmm²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	15.4	0.2%	7	8	10	289	69,000	4,166,667
A pillar	81.0	1.2%	35	33	44	611	57,255	28,583
roof front	838.5	12.8%	39	172	181	468	2,08,289	15,104
roof rear	1065.9	16.3%	172	52	179	468	2,07,264	15,104
C upper	29.4	0.4%	18	29	34	1187	1,09,719	1,40,625
C lower	175.2	2.7%	36	13	39	289	1,18,936	83,333
Rocker rear	1430.0	21.7%	6	120	123	289	9,99,145	4,166,667
Rocker front	1583.4	24.2%	110	9	114	289	9,16,874	4,166,667
B lower	16.3	0.2%	14	2	23	289	79,758	2,10,938
B upper	1.1	0.0%	5	2	12	1187	9,383	52,083
Total beams		79.9%						
hinge-rocker	61.3	0.9%						
hinge-A	71.4	1.1%						
A-roof	334.2	5.1%						
roof-C	556.4	8.5%						
C up-Clower	233.6	3.6%						
C-rocker	33.5	0.5%						
B-rocker	25.5	0.4%						
B-roof	0.8	0.0%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		20.1%						
Grand total	6543.0	100.0%						

Figure 14 Results of Initial dimensions

Iteration 1

Material	E N/mm²	G N/mm²	ρ kg/mm³
	207000	78000	7.830E-06

mm

Roof Rail h=	25	Izz=	15104.2	mm⁴
w=	40	A=	130	mm²
t=	1.0	mass=	1.27	kg

B-Upper h=	50	Izz=	52083.3	
w=	25	A=	150	
t=	1.0	mass=	0.59	

Kj= 3.80E+06 Nmm/rad

Kj= 3.80E+06 Nmm/rad

Kj= 3.80E+06 Nmm/rad

A Pillar h=	35	Izz=	28583.3	
w=	35	A=	140	
t=	1.0	mass=	0.78	

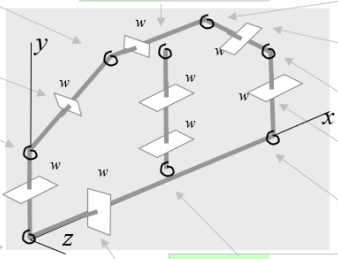
C-Upper h=	75	Izz=	140625.0	
w=	25	A=	200	
t=	1.0	mass=	0.88	

Kj= 2.80E+07 Nmm/rad

Kj= 2.80E+07 Nmm/rad

Hinge Pillar h=	100	Izz=	416666.7	
w=	50	A=	300	
t=	1.0	mass=	1.76	

C-Lower h=	50	Izz=	83333.3	
w=	50	A=	200	
t=	1.0	mass=	1.17	



Kj= 3.50E+07 Nmm/rad

Kj= 1.29E+08 Nmm/rad

Kj= 3.50E+07 Nmm/rad

Rocker h=	125	Izz=	794270.8	
w=	60	A=	370	
t=	1.0	mass=	5.79	

B-Lower h=	75	Izz=	210937.5	
w=	50	A=	250	
t=	1.0	mass=	1.47	

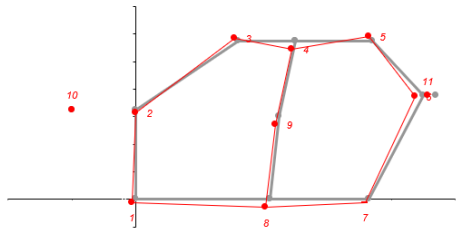
w is width into screen (mm)
h is section height (mm)
t is thickness (mm)

Side frame beam mass= 13.71 kg
(one side)

Figure 15 Iteration 1

Back to Geometry

Side Frame Results-Deflections, Energy, Stress



Deflection scale 20

	Deflections (mm) x	y
1	-1.58	-1.21
2	0.00	-1.22
3	-1.31	0.86
4	-1.33	-3.04
5	-1.35	1.73
6	-3.32	-0.25
7	-1.55	-1.23
8	-1.56	-3.02
9	-1.49	-3.02
10	0.00	0.00
11	-3.32	0.00

	Bending Energy Nmm	bending energy %	Bending Stress Nmm² end 1	end 2	total stress Nmm²	Cap buckling stress Nmm²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	7.9	0.2%	0	9	11	299	78,134	4,16,667
A pillar	97.3	2.0%	41	31	50	611	67,408	28,583
roof front	611.8	12.5%	37	148	157	468	1,79,260	15,104
roof rear	769.2	15.7%	146	46	153	468	1,76,513	15,104
C upper	36.6	0.7%	16	33	37	119.7	1,23,526	1,40,625
C lower	125.0	2.6%	40	3	42	299	1,32,225	83,333
Rocker rear	955.3	19.5%	0	87	89	208	11,02,594	7,94,271
Rocker front	1155.6	23.6%	83	1	85	208	10,48,352	7,94,271
B lower	6.1	0.1%	9	0	19	299	51,629	2,10,938
B upper	0.2	0.0%	0	2	8	119.7	4,760	52,083
Total beams		76.9%						
hinge-rocker	1.2	0.0%						
hinge-A	94.7	1.9%						
A-roof	299.1	6.1%						
roof-C	430.7	8.8%						
C up-Clower	292.1	6.0%						
C-rocker	0.4	0.0%						
B-rocker	10.8	0.2%						
B-roof	1.9	0.0%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		23.1%						
Grand total	4906.1	100.0%						

Figure 16 Results of Iteration 1

Iteration 2

Material	E N/mm²	G N/mm²	ρ kg/mm³
	207000	78000	7.830E-06

	mm		
Roof Rail h=	50	Izz= 70833.3	mm⁴
w=	40	A= 180	mm²
t=	1.0	mass= 1.76	kg

B-Upper h=	50	Izz= 52083.3
w=	25	A= 150
t=	1.0	mass= 0.59

Kj= 3.80E+06 Nmm/rad

Kj= 3.80E+06 Nmm/rad

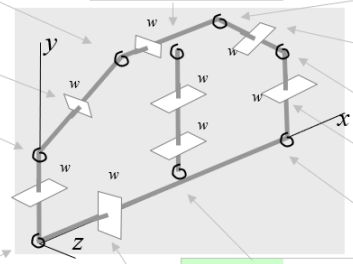
Kj= 3.80E+06 Nmm/rad

A Pillar h=	35	Izz= 28583.3
w=	35	A= 140
t=	1.0	mass= 0.78

Kj= 2.80E+07 Nmm/rad

Hinge Pillar h=	100	Izz= 416666.7
w=	50	A= 300
t=	1.0	mass= 1.76

Kj= 3.50E+07 Nmm/rad



C-Upper h=	75	Izz= 140625.0
w=	25	A= 200
t=	1.0	mass= 0.88

Kj= 2.80E+07 Nmm/rad

C-Lower h=	50	Izz= 83333.3
w=	50	A= 200
t=	1.0	mass= 1.17

Kj= 3.50E+07 Nmm/rad

Kj= 1.29E+08 Nmm/rad

Rocker h=	125	Izz= 794270.8
w=	60	A= 370
t=	1.0	mass= 5.79

B-Lower h=	75	Izz= 210937.5
w=	50	A= 250
t=	1.0	mass= 1.47

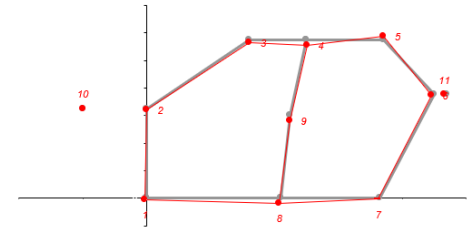
w is width into screen (mm)
h is section height (mm)
t is thickness (mm)

Side frame beam mass= 14.20 kg
(one side)

Figure 17 Iteration 2

Back to Geometry

Side Frame Results-Deflections, Energy, Stress



Deflection scale 20

	Deflections (mm) x	y
1	-0.64	-0.49
2	0.00	-0.50
3	0.29	-1.05
4	0.28	-1.91
5	0.26	1.26
6	-1.08	-0.08
7	-0.62	-0.32
8	-0.63	-1.79
9	-0.21	-1.82
10	0.00	0.00
11	-1.08	0.00

	Bending Energy Nmm	bending energy %	Bending Stress Nmm² end 1	end 2	total stress Nmm²	Cap buckling stress Nmm²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	1.7	0.1%	3	2	6	289	26,083	4,16,667
A pillar	2.9	0.1%	2	7	17	611	11,862	28,583
roof front	478.0	16.9%	2	110	117	468	3,11,453	70,833
roof rear	621.7	22.0%	113	13	118	468	3,20,857	70,833
C upper	9.2	0.3%	11	16	21	1187	60,625	1,40,625
C lower	61.4	2.2%	21	8	24	289	69,428	83,333
Rocker rear	614.7	21.7%	3	71	73	208	9,02,007	7,94,271
Rocker front	660.2	23.3%	63	3	67	208	8,05,952	7,94,271
B lower	20.0	0.7%	17	0	24	289	92,837	2,10,938
B upper	0.3	0.0%	1	3	12	1187	6,185	52,083
Total beams		87.3%						
hinge-rocker	14.9	0.5%						
hinge-A	2.0	0.1%						
A-roof	9.6	0.3%						
roof-C	200.3	7.1%						
C up-Clower	75.6	2.7%						
C-rocker	14.4	0.5%						
B-rocker	34.7	1.2%						
B-roof	8.1	0.3%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		12.7%						
Grand total	2823.6	100.0%						

Figure 18 Results of Iteration 2

Iteration 3

Material	E N/mm²	G N/mm²	ρ kg/mm³
	207000	78000	7.830E-06

Roof Rail h=	50	Izz= 241666.7	mm⁴
w=	80	A= 520	mm²
t=	2.0	mass= 5.09	kg

B-Upper h=	50	Izz= 52083.3
w=	25	A= 150
t=	1.0	mass= 0.59

Kj= 3.80E+06 Nmm/rad

Kj= 3.80E+06 Nmm/rad

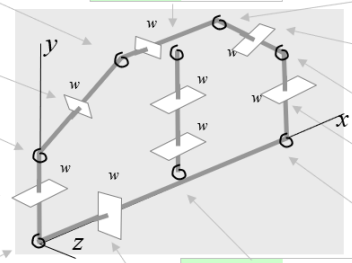
Kj= 3.80E+06 Nmm/rad

A Pillar h=	35	Izz= 28583.3
w=	35	A= 140
t=	1.0	mass= 0.78

Kj= 2.80E+07 Nmm/rad

Hinge Pillar h=	100	Izz= 416666.7
w=	50	A= 300
t=	1.0	mass= 1.76

Kj= 3.50E+07 Nmm/rad



C-Upper h=	75	Izz= 140625.0
w=	25	A= 200
t=	1.0	mass= 0.88

Kj= 2.80E+07 Nmm/rad

C-Lower h=	50	Izz= 83333.3
w=	50	A= 200
t=	1.0	mass= 1.17

Kj= 3.50E+07 Nmm/rad

Kj= 1.29E+08 Nmm/rad

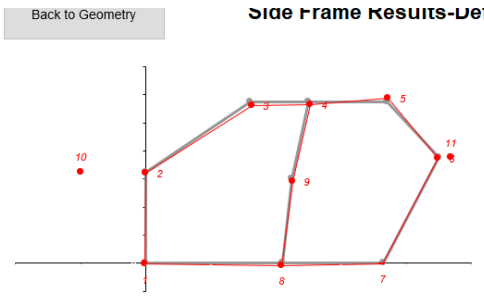
Rocker h=	125	Izz= 1588541.7
w=	60	A= 740
t=	2.0	mass= 11.59

B-Lower h=	75	Izz= 210937.5
w=	50	A= 250
t=	1.0	mass= 1.47

w is width into screen (mm)
h is section height (mm)
t is thickness (mm)

Side frame beam mass= 23.32 kg
(one side)

Figure 19 Iteration 3



Back to Geometry

Side Frame Results-Deflections, Energy, Stress

eflection scale 20

	Deflections (mm) x	y
1	-0.38	-0.29
2	0.00	-0.30
3	0.69	-1.50
4	0.68	-1.00
5	0.68	1.11
6	-0.48	-0.03
7	-0.37	-0.08
8	-0.37	-0.85
9	0.15	-0.90
10	0.00	0.00
11	-0.48	0.00

	Bending Energy Nmm	bending energy %	Bending Stress Nmm⁻² end 1	end 2	total stress Nmm⁻²	Cap buckling stress Nmm⁻²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	0.0	0.0%	1	0	3	289	5,637	4,16,667
A pillar	3.9	0.3%	8	3	19	611	13,847	28,583
roof front	181.9	13.9%	1	36	38	468	3,45,885	2,41,667
roof rear	238.0	18.2%	37	2	39	468	3,59,072	2,41,667
C upper	5.3	0.4%	8	12	18	1187	46,132	1,40,625
C lower	32.3	2.5%	16	4	20	289	54,722	83,333
Rocker rear	298.2	22.7%	1	35	36	832	8,82,128	15,88,542
Rocker front	320.1	24.4%	31	1	32	832	7,84,229	15,88,542
B lower	20.0	1.5%	17	0	24	289	94,487	2,10,938
B upper	0.9	0.1%	1	5	14	1187	9,665	52,083
Total beams		83.9%						
hinge-rocker	2.1	0.2%						
hinge-A	1.3	0.1%						
A-roof	9.9	0.8%						
roof-C	93.9	7.2%						
C up-Clower	45.5	3.5%						
C-rocker	4.7	0.4%						
B-rocker	36.0	2.7%						
B-roof	17.1	1.3%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		16.1%						
Grand total	1311.2	100.0%						

Figure 19 Results of Iteration 3

For the iterations,

we start by identifying the component with the largest strain energy (in this case, the roof rail was found first, then the rocker). And in order to enhance stiffness, we start by increasing the member's most useful dimension, first its height and then its breadth. The thickness of the members is raised if an additional stiffness increase is necessary.

Numerous rounds of expanding one dimension at a time were carried out in this precise order; however, only the three most important iterations are shown in the table below. (Note that the element dimensions were raised one after the other in the original iteration sequence).

	Initial	Iteration 1	Iteration 2	Iteration 3
Rocker	h=100mm w=50mm t=1mm	h=125mm w=60mm t=1mm	h=125mm w=60mm t=2mm	h=125mm w=60mm t=2mm
Roof rail	h=25mm w=40mm t=1mm	h=25mm w=40mm t=1mm	h=50mm w=40mm t=1mm	H=50mm w=80mm t=2mm
Deflection	-4.02mm @9	-3.02mm@9	-1.82mm@9	-0.90mm@9
Stiffness	830.22 N/mm	1105.132 N/mm	1833.79 N/mm	3708.33 N/mm
Mass	12.61kg	13.71kg	14.20kg	23.32kg

Table 2 Iterations data

Thus, the ultimate rigidity for one side is 3708.33 N/mm, which is greater than the minimum required 3500 N/mm value.

Part 3: Body Structural Torsion Stiffness Modeling and Development

Applying the discrete torsion stiffness model developed in class, integrated with a body-side torsion finite element model, to improve the torsion stiffness of the body side. The effective shear rigidity data for 6 of the 7-panel components are listed. Assuming that a shear load, Q4, applied to the side frame is 2678 N. This results from a total torsion vehicle load of 7,730,000 N-mm.

We conduct a body-side torsion stiffness evaluation to determine (Gt)EFF using the torsion finite element model configuration covered in class and accessible in the Excel-based software package. (Using the thickness and cross-sectional values for the body side beams from Part 2).

$$K = \frac{1}{\left(\frac{q}{T}\right)^2 \sum_{\text{ALL SURFACES}} \left[\frac{\text{area of surface i}}{(Gt)_{EFF}}\right]_{\text{SURF i}}}$$

Panel	Area of panel (mm^2)	Effective shear rigidity $(Gt)_{EFF}$ (N/mm)	$\left[\frac{\text{area of surface i}}{(Gt)_{EFF\,i}}\right]$ (mm^3/N)
dash	1170000	80000	14.6
windshield	1103087	80000	13.8
roof	1950000	80000	24.4
back light	872067	80000	10.9
seat back	1170000	80000	14.6
floor	3120000	80000	39.0

$$(Gt)_{EFF} = \frac{F*H}{\delta*L}$$

$$(Gt)_{EFF} = \frac{2678*1150}{22.40*2250}$$

$$(Gt)_{EFF} = 61.105 \text{ N/mm}$$

Material	E	G	ρ
	N/mm²	N/mm²	kg/mm³
	207000	78000	7.830E-06

	mm		
Roof Rail h=	50	Izz= 241666.7	mm⁴
w=	80	A= 520	mm²
t=	2.0	mass= 5.09	kg

B-Upper h=	50	Izz= 52083.3
w=	25	A= 150
t=	1.0	mass= 0.59

Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad

A Pillar h=	35	Izz= 28583.3
w=	35	A= 140
t=	1.0	mass= 0.78

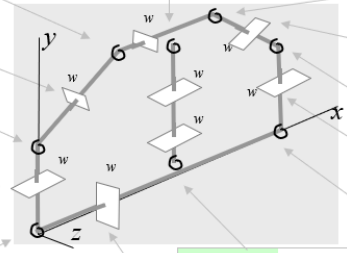
C-Upper h=	75	Izz= 140625.0
w=	25	A= 200
t=	1.0	mass= 0.88

Kj=	2.80E+07	Nmm/rad
Kj=	2.80E+07	Nmm/rad

Hinge Pillar h=	100	Izz= 416666.7
w=	50	A= 300
t=	1.0	mass= 1.76

C-Lower h=	50	Izz= 83333.3
w=	50	A= 200
t=	1.0	mass= 1.17

Kj=	3.50E+07	Nmm/rad
Kj=	1.29E+08	Nmm/rad
Kj=	3.50E+07	Nmm/rad



Rocker h=	125	Izz= 1588541.7
w=	60	A= 740
t=	2.0	mass= 11.59

B-Lower h=	75	Izz= 210937.5
w=	50	A= 250
t=	1.0	mass= 1.47

w is width into screen (mm)
h is section height (mm)
t is thickness (mm)
mass in Kg

Side frame beam mass= 23.32 kg
(one side)

Figure 20 Initial Values

Back to Geometry

Recalculate

Results

Side Frame Restrains and Loads

Node Restraints

Node	X	Y	Rotation
1	F	F	
7		F	

Add node number and F to fix the deflection

Loads (N)

Beam	Position	Direction	Magnitude
4	0	x	2678

Add beam number, position of load, direction (X or Y)
Position=0 at beginning of beam (base of arrow on diagram)
Add magnitude (may be negative)

For bending test use

10	F	F
11		F

For effective shear rigidity use

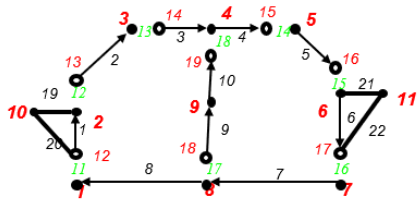
1	F	F
7		F

For bending test use

9	0	Y	magnitude

For effective shear rigidity use

4	0	X	magnitude



numbers in black=beams
numbers in bold red=primary nodes
numbers in green=joints
numbers in small red=secondary nodes

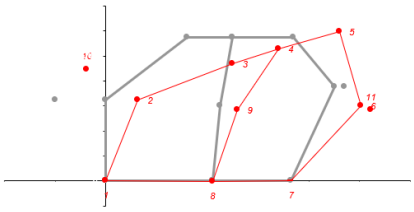
Apply loads to beams
Apply restraints to nodes

Used for joint stiffness connections

Figure 21 Loads and Restrains

Back to Geometry

Side Frame Results-Deflections, Energy, Stress



Deflection scale 20

	Deflections (mm) x	y
1	0.00	0.00
2	15.65	0.01
3	22.40	-10.54
4	22.40	-4.68
5	22.40	2.11
6	12.92	-7.38
7	0.01	0.00
8	0.01	-0.38
9	8.53	-1.47
10	15.65	12.13
11	12.92	-9.12

	Bending Energy Nmm	bending energy %	Bending Stress end 1 Nmm⁻²	end 2	total stress Nmm⁻²	Cap buckling stress Nmm⁻²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	602.4	2.0%	89	12	92	298	7,416,531	4,16,667
A pillar	718.2	2.4%	68	48	79	611	1,10,780	28,503
roof front	108.9	0.4%	7	31	34	468	3,00,835	2,41,667
roof rear	112.2	0.4%	15	13	19	468	1,46,726	2,41,667
C upper	28.6	0.1%	34	16	36	1187	1,28,629	1,40,625
C lower	1447.9	4.9%	19	115	116	298	3,83,717	83,333
Rocker rear	264.7	0.9%	15	37	38	832	9,42,969	15,68,542
Rocker front	259.5	0.9%	5	30	32	832	7,54,747	15,68,542
B lower	4166.0	14.0%	189	82	192	298	10,65,408	2,10,938
B upper	1416.2	4.7%	220	70	225	1187	4,59,157	52,083
Total beams		30.6%						
hinge-rocker	7897.2	26.8%						
hinge-A	194.2	0.7%						
A-roof	720.3	2.4%						
roof-C	2182.1	7.3%						
C up-Clower	67.0	0.2%						
C-rocker	2128.3	7.1%						
B-rocker	4454.8	14.9%						
B-roof	2942.8	9.8%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.1	0.0%						
rear end diag	0.0	0.0%						
Total joints		69.4%						
Grand total	29832.6	100.0%						

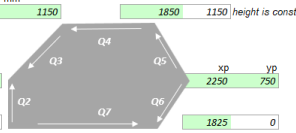
Figure 22 Initial Results

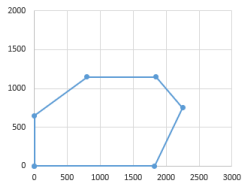
SEDAN-Rear torque applied to seat back

Applied torque 77,30,000 Nmm

Cabin width 1550 mm

Coordinates mm: 800 1150; 1850 1150 *height is constrained to be horizontal*; 0 650; 0 0; 1825 0; xp 2250 yp 750





A

Q1	Q2	Q3	Q4	Q5	Q6	Q7
650	1550	0	0	0	0	0
943	0	-1550	0	0	0	0
1050	0	0	-1550	0	0	0
566	0	0	0	-1550	0	0
862	0	0	0	0	1550	0
1825	0	0	0	0	0	-1550
0	2250	-1108	-400	0	0	-750

T

	Source of equation
7730000	Dash
0	Windshield
0	Roof
0	Backlite
7730000	Rear
0	Floor
0	Sideframe

From sum of moments about point p (Clockwise positive)

Area of sideframe= 2148125 mm²

A^{-1}

0.000523714	0.000257827	0.000093105	0.000000000	0.000000000	0.000174571	-0.000360781
0.000425539	-0.000108121	-0.000039044	0.000000000	0.000000000	-0.000073207	0.000151295
0.000318755	-0.000488237	0.000056668	0.000000000	0.000000000	0.000106252	-0.000219587
0.000354774	0.000174657	-0.000582090	0.000000000	0.000000000	0.000118258	-0.000244400
0.000191134	0.000094096	0.000033979	-0.000645161	0.000000000	0.000063711	-0.000131670
-0.000291268	-0.000143393	-0.000051781	0.000000000	0.000645161	-0.000097089	0.000200652
0.000616631	0.000303570	0.000109623	0.000000000	0.000000000	-0.000439618	-0.000424790

x

T

7730000
0
0
0
7730000
0
0

=

	Q
Q1	4048
Q2	3289
Q3	2464
Q4	2742
Q5	1477
Q6	2736
Q7	4767

	q
Q1	2.612
Q2	5.061
Q3	2.612
Q4	2.612
Q5	2.612
Q6	3.173
Q7	2.612

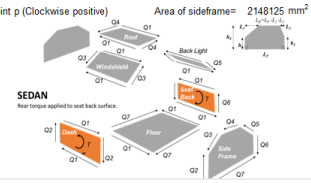


Figure 23 Input Values for Torsional stiffness

SEDAN-Rear torque applied to seat back

Uses cabin dimensions and applied torque input on sheet Cabin Shear Flow Under Torsion

Cabin widt 1550 mm

Panel	Size	Area of Panel A (mm²)	Effective Shear Rigidity *Gt* (N/mm)	Shear Flexibility Contribution A/Gt (mm³/N)	Pecent Contribution to Shear Flexibility
Dash	650 mm	1.01E+06	80,000	12.6	0.0%
Winshield	943 mm	1.46E+06	80,000	18.3	0.0%
Roof	1050 mm	1.63E+06	80,000	20.3	0.0%
Back Light	566 mm	8.77E+05	80,000	11.0	0.0%
Seat Back	862 mm	1.34E+06	80,000	16.7	0.0%
Floor	1825 mm	2.83E+06	80,000	35.4	0.1%
Side Frame-Left		2.15E+06	61	35215.2	49.9%
Side Frame-Right		2.15E+06	61	35215.2	49.9%
			sum	70544.6	100.0%

Angle of twist	6.23E-02	rad	*under T=*	77,30,000 Nmm
Torsional stiffness	1.24E+08	Nmm/rad		
Torsional stiffness	2,167	Nm/deg		

Figure 24 Initial value of Torsional stiffness

Material	E	G	ρ
	N/mm²	N/mm²	kg/mm³
	207000	78000	7.830E-06

	mm		
Roof Rail h=	50	Izz= 241666.7	mm⁴
w=	80	A= 520	mm²
t=	2.0	mass= 5.09	kg

B-Upper h=	50	Izz= 52083.3	
w=	25	A= 150	
t=	1.0	mass= 0.59	

Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad

A Pillar h=	35	Izz= 28583.3	
w=	35	A= 140	
t=	1.0	mass= 0.78	

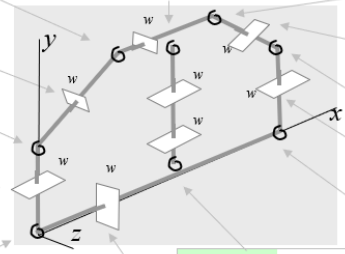
C-Upper h=	75	Izz= 140625.0	
w=	25	A= 200	
t=	1.0	mass= 0.88	

Kj=	2.80E+07	Nmm/rad
Kj=	2.80E+07	Nmm/rad

Hinge Pillar h=	100	Izz= 416666.7	
w=	50	A= 300	
t=	1.0	mass= 1.76	

C-Lower h=	50	Izz= 83333.3	
w=	50	A= 200	
t=	1.0	mass= 1.17	

Kj=	3.50E+07	Nmm/rad
Kj=	1.29E+08	Nmm/rad
Kj=	3.50E+07	Nmm/rad



Rocker h=	125	Izz= 1588541.7	
w=	60	A= 740	
t=	2.0	mass= 11.59	

B-Lower h=	150	Izz= 5062500.0	
w=	100	A= 1500	
t=	3.0	mass= 8.81	

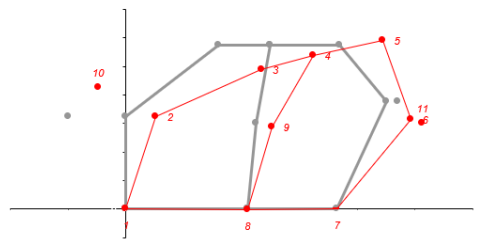
w is width into screen (mm)
h is section height (mm)
t is thickness (mm)
mass in Kg

Side frame beam mass= 30.66 kg
(one side)

Figure 25 Iteration 1

Back to Geometry

Side Frame Results-Deflections, Energy, Stress



Deflection scale 20

	Deflections (mm) x	y
1	0.00	0.00
2	12.96	0.01
3	18.47	-8.60
4	18.48	-3.81
5	18.48	1.72
6	10.67	-6.09
7	0.01	0.00
8	0.01	-0.26
9	6.86	-1.13
10	12.96	10.05
11	10.67	-7.53

	Bending Energy Nmm	bending energy %	Bending Stress end 1 Nmm²	end 2	total stress Nmm²	Cap buckling stress Nmm²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	421.2	1.7%	74	10	77	299	6,20,615	4,16,667
A pillar	501.2	2.0%	58	39	67	611	93,994	28,583
roof front	79.3	0.3%	6	26	29	468	2,55,744	2,41,667
roof rear	74.2	0.3%	12	11	17	468	1,18,389	2,41,667
C upper	26.8	0.1%	28	13	29	1197	1,05,592	1,40,625
C lower	986.1	4.0%	15	95	96	299	3,17,209	83,333
Rocker rear	309.5	1.3%	13	39	40	832	10,03,432	15,88,542
Rocker front	162.5	0.7%	16	25	27	832	6,31,611	15,88,542
B lower	298.7	1.2%	21	9	21	674	13,85,338	50,62,500
B upper	2745.1	11.2%	297	60	300	1197	6,18,154	52,083
Total beams		22.8%						
hinge-rocker	5600.4	22.8%						
hinge-A	140.0	0.6%						
A-roof	492.8	2.0%						
roof-C	1470.0	6.0%						
C up-Clower	44.0	0.2%						
C-rocker	1454.4	5.9%						
B-rocker	7532.8	30.6%						
B-roof	2270.3	9.2%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		77.2%						
Grand total	24599.3	100.0%						

Figure 26 Results of Iteration 1

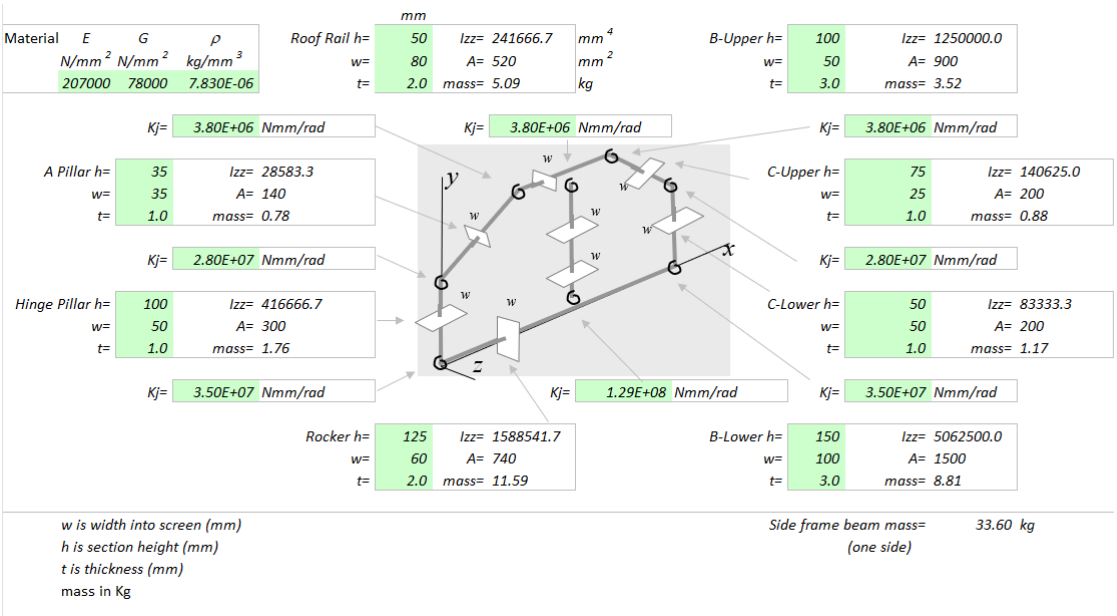
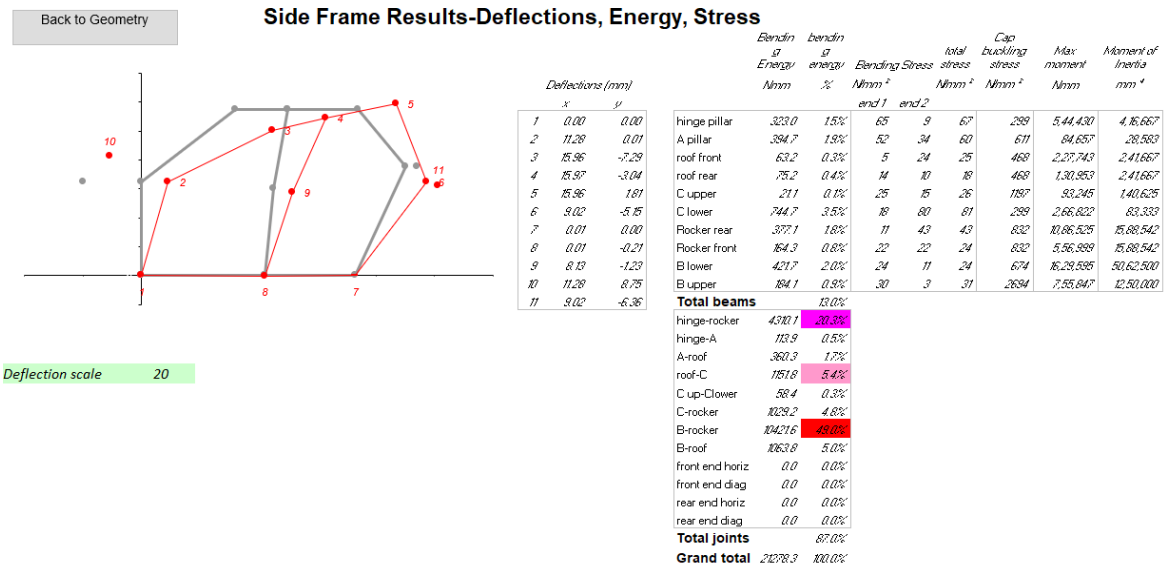


Figure 27 Iteration 2



	Deflections (mm) x	y
1	0.00	0.00
2	11.28	0.01
3	15.96	-7.29
4	15.97	-3.04
5	15.96	1.61
6	9.02	-5.15
7	0.01	0.00
8	0.01	-0.21
9	8.13	-1.23
10	11.28	8.75
11	9.02	-6.36

	Bending Energy Nmm	bending energy %	Bending Stress Nmm 2 end 1	end 2	total stress Nmm 2	Cap buckling stress Nmm 2	Max moment Nmm	Moment of Inertia mm 4
hinge pillar	323.0	1.5%	65	9	67	299	5,44,430	4,16,667
A pillar	394.7	1.9%	52	34	60	611	84,657	28,583
roof front	63.2	0.3%	5	24	25	468	2,27,743	2,41,667
roof rear	175.2	0.4%	14	10	18	468	1,30,953	2,41,667
C upper	21.1	0.1%	25	15	26	1197	93,245	1,40,625
C lower	744.7	3.5%	18	80	81	299	2,66,822	83,333
Rocker rear	377.1	1.8%	11	43	43	832	10,86,525	15,88,542
Rocker front	164.3	0.8%	22	22	24	832	5,56,999	15,88,542
B lower	421.7	2.0%	24	11	24	674	16,29,595	50,62,500
B upper	194.1	0.9%	30	3	31	2694	7,55,847	12,50,000
Total beams		13.0%						
hinge-rocker	4,910.1	23.3%						
hinge-A	113.9	0.5%						
A-roof	360.3	1.7%						
roof-C	1151.8	5.4%						
C up-Clower	58.4	0.3%						
C-rocker	1029.2	4.8%						
B-rocker	10421.6	49.0%						
B-roof	1063.8	5.0%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		87.0%						
Grand total	21278.3	100.0%						

Figure 28 Results of Iteration 2

Material	E	G	ρ
	N/mm²	N/mm²	kg/mm³
	207000	78000	7.830E-06

	mm		
Roof Rail h=	50	Izz= 362500.0	mm⁴
w=	80	A= 780	mm²
t=	3.0	mass= 7.63	kg

B-Upper h=	100	Izz= 1250000.0	
w=	50	A= 900	
t=	3.0	mass= 3.52	

Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad

A Pillar h=	70	Izz= 686000.0	
w=	70	A= 840	
t=	3.0	mass= 4.65	

C-Upper h=	150	Izz= 3375000.0	
w=	50	A= 1200	
t=	3.0	mass= 5.25	

Kj=	2.80E+07	Nmm/rad
Kj=	2.80E+07	Nmm/rad

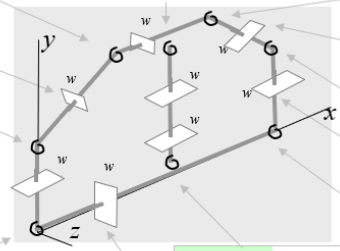
Hinge Pillar h=	200	Izz= 10000000.0	
w=	100	A= 1800	
t=	3.0	mass= 10.57	

C-Lower h=	100	Izz= 2000000.0	
w=	100	A= 1200	
t=	3.0	mass= 7.05	

Kj=	3.50E+07	Nmm/rad
Kj=	1.29E+08	Nmm/rad
Kj=	3.50E+07	Nmm/rad

Rocker h=	125	Izz= 1588541.7	
w=	60	A= 740	
t=	2.0	mass= 11.59	

B-Lower h=	150	Izz= 5062500.0	
w=	100	A= 1500	
t=	3.0	mass= 8.81	



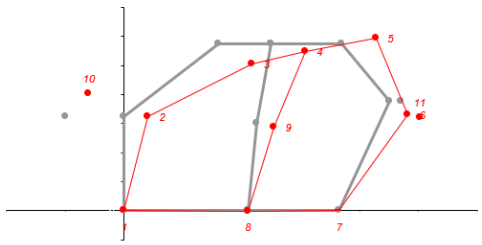
w is width into screen (mm)
h is section height (mm)
t is thickness (mm)
mass in Kg

Side frame beam mass= 59.07 kg
(one side)

Figure 29 Iteration 3

Back to Geometry

Side Frame Results-Deflections, Energy, Stress



	Deflections (mm) x	y
1	0.00	0.00
2	9.99	0.00
3	14.33	-6.67
4	14.33	-2.79
5	14.33	1.93
6	7.91	-4.51
7	0.01	0.00
8	0.01	-0.25
9	7.29	-1.17
10	9.99	7.74
11	7.91	-5.57

	Bending Energy Nmm	bending energy %	Bending Stress Nmm² end 1	end 2	total stress Nmm²	Cap buckling stress Nmm²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	10.6	0.1%	5	2	6	674	5,13,630	1,00,00,000
A pillar	57.5	0.3%	9	4	11	1375	1,84,600	6,86,000
roof front	67.8	0.4%	4	20	21	1052	2,89,078	3,62,500
roof rear	87.0	0.5%	14	6	17	1052	1,99,938	3,62,500
C upper	2.1	0.0%	2	3	3	2694	1,44,500	33,75,000
C lower	74.4	0.4%	4	9	10	674	3,72,955	20,00,000
Rocker rear	349.9	1.8%	15	42	43	832	10,73,990	15,88,542
Rocker front	121.0	0.6%	16	21	23	832	5,24,575	15,88,542
B lower	345.3	1.8%	22	10	22	674	14,69,198	50,62,500
B upper	153.4	0.8%	28	3	28	2694	6,90,770	12,50,000
Total beams		6.6%						
hinge-rocker	3851.2	20.1%						
hinge-A	573.8	3.0%						
A-roof	591.2	3.1%						
roof-C	1065.4	5.6%						
C up-Clower	383.5	2.0%						
C-rocker	2009.0	10.5%						
B-rocker	8467.5	44.3%						
B-roof	911.7	4.8%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		93.4%						
Grand total	19122.2	100.0%						

Deflection scale 20

Figure 30 Results of Iteration 3

	Initial	Iteration 1	Iteration 2	Iteration 3
B Lower	h=75mm w=50mm t=1mm	h=150mm w=100mm t=3mm	h=150mm w=100mm t=3mm	h=150mm w=100mm t=3mm
B Upper	h=50mm w=25mm t=1mm	h=50mm w=25mm t=1mm	h=100mm w=50mm t=3mm	H=100mm w=50mm t=3mm
All beams	-	-	-	Maximum allowable dimensions
Deflection	22.40mm @4	18.48@4	15.97@4	14.33@4
Mass	23.32 kg	30.66kg	33.60kg	59.07 kg
Stiffness	2167 Nm/deg			3390 Nm/deg

Table 3 Iterations Data

SEDAN-Rear torque applied to seat back

Uses cabin dimensions and applied torque input on sheet Cabin Shear Flow Under Torsion

Cabin widt *1550 mm*

Panel	Size	Area of Panel A (mm^2)	Effective Shear Rigidity *Gt* (N/mm)	Shear Flexibility Contribution A/Gt (mm^3/N)	Pecent Contribution to Shear Flexibility
Dash	*650 mm*	*1.01E+06*	*80,000*	*12.6*	*0.0%*
Winshield	*943 mm*	*1.46E+06*	*80,000*	*18.3*	*0.0%*
Roof	*1050 mm*	*1.63E+06*	*80,000*	*20.3*	*0.0%*
Back Light	*566 mm*	*8.77E+05*	*80,000*	*11.0*	*0.0%*
Seat Back	*862 mm*	*1.34E+06*	*80,000*	*16.7*	*0.0%*
Floor	*1825 mm*	*2.83E+06*	*80,000*	*35.4*	*0.1%*
Side Frame-Left		*2.15E+06*	*96*	*22491.1*	*49.9%*
Side Frame-Right		*2.15E+06*	*96*	*22491.1*	*49.9%*
			sum	*45096.4*	*100.0%*

Angle of twist	*3.98E-02*	rad	*under T=*	*77,30,000 Nmm*
Torsional stiffness	*1.94E+08*	Nmm/rad		
Torsional stiffness	*3,390*	Nm/deg		

Figure 31 Torsional stiffness value for Iteration 3

We increased the beams' size to their maximum permitted dimensions in the third iteration. The torsional stiffness value was 3390 Nm/deg. However, the maximum deflection was still 14.33mm. As a result, we chose to double the stiffness of the joint G and raise it sequentially up to a maximum of 10. The iterations were then carried out once more using the modified joint G stiffness value of 2.57 x 109 Nmm/rad.

Back to Geometry | Enter Loads and restraints

Side Frame Sections and Joint Properties

Material	E	G	ρ
	N/mm²	N/mm²	kg/mm³
	207000	78000	7.830E-06

	mm		
Roof Rail h=	50	Izz= 241666.7	mm⁴
w=	80	A= 520	mm²
t=	2.0	mass= 5.09	kg

B-Upper h=	50	Izz= 52083.3
w=	25	A= 150
t=	1.0	mass= 0.59

Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad
Kj=	3.80E+06	Nmm/rad

A Pillar h=	35	Izz= 28583.3
w=	35	A= 140
t=	1.0	mass= 0.78

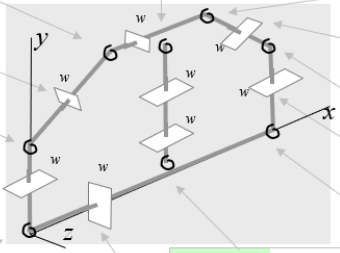
C-Upper h=	75	Izz= 140625.0
w=	25	A= 200
t=	1.0	mass= 0.88

Kj=	2.80E+07	Nmm/rad
Kj=	2.80E+07	Nmm/rad

Hinge Pillar h=	100	Izz= 416666.7
w=	50	A= 300
t=	1.0	mass= 1.76

C-Lower h=	50	Izz= 83333.3
w=	50	A= 200
t=	1.0	mass= 1.17

Kj=	3.50E+07	Nmm/rad
Kj=	2.57E+09	Nmm/rad
Kj=	3.50E+07	Nmm/rad



Rocker h=	125	Izz= 1588541.7
w=	60	A= 740
t=	2.0	mass= 11.59

B-Lower h=	150	Izz= 5062500.0
w=	100	A= 1500
t=	3.0	mass= 8.81

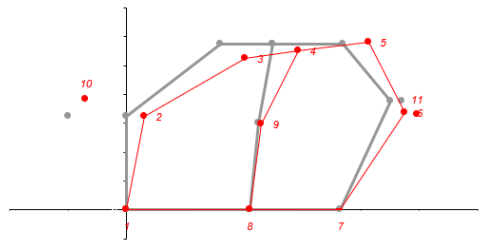
w is width into screen (mm)
h is section height (mm)
t is thickness (mm)
mass in Kg

Side frame beam mass= 30.66 kg
(one side)

Figure 32 Iteration 1 (after changing stiffness value)

Back to Geometry

Side Frame Results-Deflections, Energy, Stress



Deflection scale 20

	Deflections (mm) x	y
1	0.00	0.00
2	7.57	0.00
3	10.75	-4.96
4	10.75	-2.33
5	10.75	0.72
6	6.39	-3.64
7	0.01	0.00
8	0.01	0.00
9	1.25	-0.16
10	7.57	5.67
11	6.39	-4.50

	Bending Energy Nmm	bending energy %	Bending Stress Nmm² end 1	end 2	total stress Nmm²	Cap buckling stress Nmm²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	156.6	1.1%	45	6	47	299	3,78,050	4,16,667
A pillar	171.9	1.2%	35	22	40	611	56,438	28,583
roof front	31.5	0.2%	3	17	18	468	1,59,815	2,41,667
roof rear	14.8	0.1%	4	6	11	468	58,620	2,41,667
C upper	9.0	0.1%	16	4	17	1197	58,927	1,40,625
C lower	322.3	2.3%	5	58	58	299	1,91,775	83,333
Rocker rear	389.1	2.8%	8	43	43	832	10,90,829	15,88,542
Rocker front	348.9	2.5%	37	15	39	832	9,30,362	15,88,542
B lower	632.8	4.4%	30	14	30	674	20,03,288	50,62,500
B upper	6494.5	45.6%	440	48	440	1197	9,16,617	52,083
Total beams		60.3%						
hinge-rocker	2078.1	14.6%						
hinge-A	50.4	0.4%						
A-roof	150.3	1.1%						
roof-C	454.6	3.2%						
C up-Clower	4.5	0.0%						
C-rocker	531.4	3.7%						
B-rocker	787.8	5.5%						
B-roof	1588.6	11.2%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		39.7%						
Grand total	14237.1	100.0%						

Figure 33 Results of Iteration 1

Back to Geometry | Enter Loads and restraints

Side Frame Sections and Joint Properties

Material	E	G	ρ
	N/mm²	N/mm²	kg/mm³
	207000	78000	7.830E-06

	mm			
Roof Rail h=	50	Izz= 241666.7	mm⁴	
w=	80	A= 520	mm²	
t=	2.0	mass= 5.09	kg	

B-Upper h=	100	Izz= 1250000.0		
w=	50	A= 900		
t=	3.0	mass= 3.52		

Kj= 3.80E+06 Nmm/rad

Kj= 3.80E+06 Nmm/rad

Kj= 3.80E+06 Nmm/rad

A Pillar h=	35	Izz= 28583.3		
w=	35	A= 140		
t=	1.0	mass= 0.78		

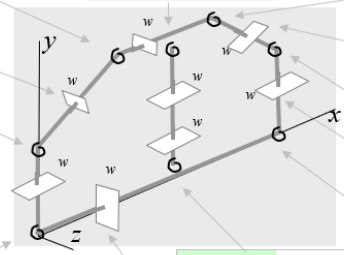
C-Upper h=	75	Izz= 140625.0		
w=	25	A= 200		
t=	1.0	mass= 0.88		

Kj= 2.80E+07 Nmm/rad

Kj= 2.80E+07 Nmm/rad

Hinge Pillar h=	100	Izz= 416666.7		
w=	50	A= 300		
t=	1.0	mass= 1.76		

C-Lower h=	50	Izz= 83333.3		
w=	50	A= 200		
t=	1.0	mass= 1.17		



Kj= 3.50E+07 Nmm/rad

Kj= 2.57E+09 Nmm/rad

Kj= 3.50E+07 Nmm/rad

Rocker h=	125	Izz= 1588541.7		
w=	60	A= 740		
t=	2.0	mass= 11.59		

B-Lower h=	150	Izz= 5062500.0		
w=	100	A= 1500		
t=	3.0	mass= 8.81		

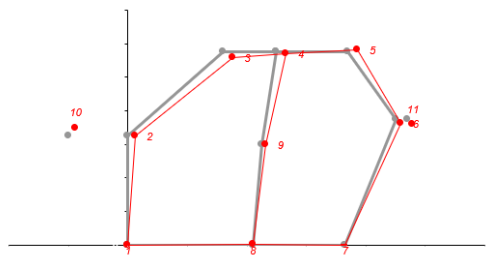
w is width into screen (mm)
h is section height (mm)
t is thickness (mm)
mass in Kg

Side frame beam mass= 33.60 kg
(one side)

Figure 34 Iteration 2 (after changing stiffness value)

Back to Geometry

Side Frame Results-Deflections, Energy, Stress



Deflection scale 20

	Deflections (mm) x	y
1	0.00	0.00
2	3.00	0.00
3	4.06	-1.64
4	4.07	-0.57
5	4.07	0.58
6	2.23	-1.26
7	0.02	0.00
8	0.02	0.18
9	1.74	-0.04
10	3.00	2.33
11	2.23	-1.56

	Bending Energy Nmm	bending energy %	Bending Stress Nmm² end 1	end 2	total stress Nmm²	Cap buckling stress Nmm²	Max moment Nmm	Moment of Inertia mm⁴
hinge pillar	32.0	0.6%	21	3	21	299	1,71,669	4,16,667
A pillar	34.1	0.6%	17	8	20	611	27,653	28,583
roof front	8.2	0.2%	1	8	9	468	79,831	2,41,667
roof rear	6.7	0.1%	4	2	9	468	42,270	2,41,667
C upper	1.5	0.0%	6	5	7	1197	23,928	1,40,625
C lower	50.3	0.9%	6	20	20	299	65,976	83,333
Rocker rear	597.0	11.2%	3	50	50	832	12,64,867	15,88,542
Rocker front	902.3	16.9%	54	7	58	832	13,80,643	15,88,542
B lower	1108.7	20.8%	39	19	39	674	26,22,591	50,62,500
B upper	553.7	10.4%	50	1	50	2694	12,52,600	12,50,000
Total beams		61.6%						
hinge-rocker	428.6	8.0%						
hinge-A	12.2	0.2%						
A-roof	19.8	0.4%						
roof-C	75.7	1.4%						
C up-Clower	7.0	0.1%						
C-rocker	62.9	1.2%						
B-rocker	1348.9	25.2%						
B-roof	95.4	1.8%						
front end horiz	0.0	0.0%						
front end diag	0.0	0.0%						
rear end horiz	0.0	0.0%						
rear end diag	0.0	0.0%						
Total joints		38.4%						
Grand total	5346.0	100.0%						

Figure 35 Results of Iteration 2

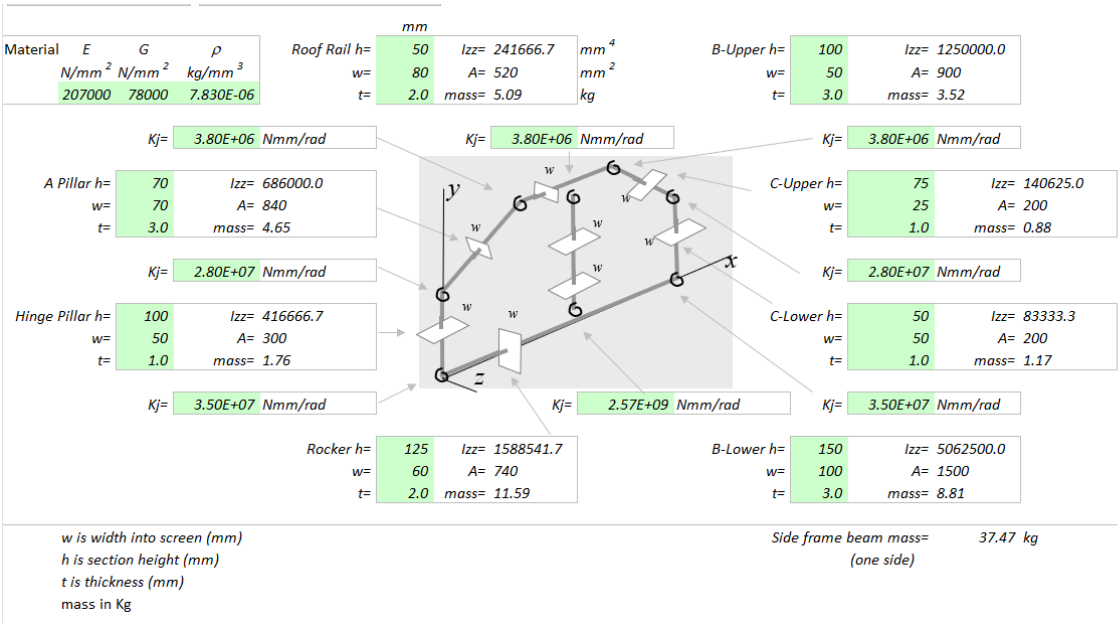


Figure 36 Iteration 3 (after changing stiffness value)

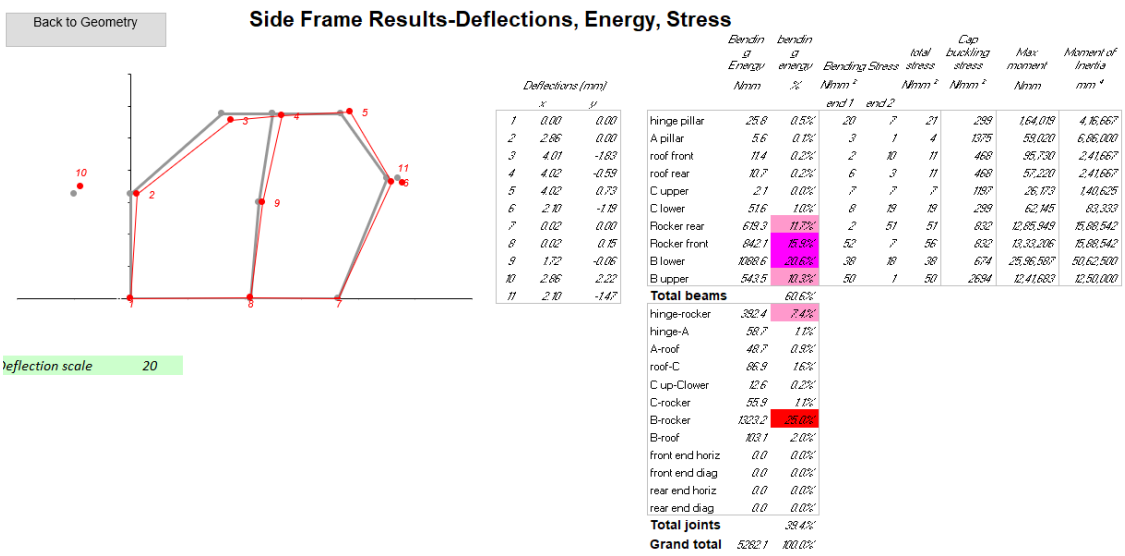


Figure 37 Results of Iteration 3

	Initial	Iteration 1	Iteration 2	Iteration 3
B Lower	h=75mm w=50mm t=1mm	h=150mm w=100mm t=3mm	h=150mm w=100mm t=3mm	h=150mm w=100mm t=3mm
B Upper	h=50mm w=25mm t=1mm	h=50mm w=25mm t=1mm	h=100mm w=50mm t=3mm	H=100mm w=50mm t=3mm
Deflection		10.75@4	4.07@4	4.02@4
Mass		30.66kg	33.60kg	37.47 kg
Stiffness				12,006 Nm/deg

Table 4 Iteration data (after changing the stiffness)

Once more, in this case, to enhance stiffness, we start by increasing the most practical dimension: first, the members' height, then their breadth. If more stiffness improvement is needed, the members' thickness is raised.

We did several iterations, adding one dimension at a time in this precise order; however, the tables only show the top three iterations. (Note that the element dimensions were raised one after the other in the original iteration sequence).

Finally,

At a deflection of 4.02 mm,

$$(Gt)_{EFF} = \frac{F*H}{\delta*L}$$

$$(Gt)_{EFF} = \frac{2678*1150}{4.02*2250}$$

$$(Gt)_{EFF} = 340.48 \text{ N/mm}$$

SEDAN-Rear torque applied to seat back

Uses cabin dimensions and applied torque input on sheet Cabin Shear Flow Under Torsion

Cabin widt 1550 mm

Panel	Size	Area of Panel A (mm^2)	Effective Shear Rigidity *Gt* (N/mm)	Shear Flexibility Contribution A/Gt (mm^3/N)	Pecent Contribution to Shear Flexibility
Dash	650 mm	1.01E+06	80,000	12.6	0.1%
Winshield	943 mm	1.46E+06	80,000	18.3	0.1%
Roof	1050 mm	1.63E+06	80,000	20.3	0.2%
Back Light	566 mm	8.77E+05	80,000	11.0	0.1%
Seat Back	862 mm	1.34E+06	80,000	16.7	0.1%
Floor	1825 mm	2.83E+06	80,000	35.4	0.3%
Side Frame-Left		2.15E+06	340	6309.1	49.6%
Side Frame-Right		2.15E+06	340	6309.1	49.6%
			sum	12732.5	100.0%

Angle of twist	1.12E-02	rad	under T=	77,30,000 Nmm
Torsional stiffness	6.88E+08	Nmm/rad		
Torsional stiffness	12,006	Nm/deg		

Figure 38 Torsional stiffness (final Iteration)

We reach the desired requirement with a torsional stiffness of 12,006 Nm/deg after the last iteration.

Part 4: Design for Crashworthiness

Part 4A: Development of Front Barrier Requirements

Develop the design for the front mid- and upper rails for the vehicle to absorb a sufficient amount of energy when it impacts a barrier at a velocity, V_o = 50 km/h. The maximum deceleration level is assumed to be a_{max} = 20 g. The design crush efficiency is η = 0.8.

- Compute the required crushable space, Δ.
- Determine the mean crush force which must be generated by the vehicle, F_{AVE}.

Given:

$V_0 = 50 \text{ km/h} = \frac{5}{18} x50 = 13.88 \text{m/sec}$

$a_{max} = 20 \text{ g} = 20 \text{ x } 9.81 = 196.2 \text{m/sec}$

As we know that,

$$\Delta = \frac{{V_o}^2}{2*\eta*a_{max}} = \frac{13.88^2}{2*0.8*196.2}$$

$\Delta = 0.613$

$F_{AVE} = \frac{m*V_o^2}{2*\Delta}$

$F_{AVE} = \frac{1911.03*13.88^2}{2*0.613}$

F_{AVE} = 300.30 KN

a) Design of the Front Mid-Rail for Energy Absorption

Assume the two vehicle mid-rails will provide 50% of the required average crumpling force, FAVE. The material is mild steel with σy = 220 N/mm2. For a width to thickness ratio, b/t = 65, determine the size of a square section required to generate the required reactive force. Compute the mass of the two rails.

If an octagonal cross-section is being considered as a design alternative, compute the size of this section needed to react the impact force. That is the resulting mass of the two mid-rails for this design?

<u>1.For Square Cross-section:</u>

As mentioned in the question,
The front mid rails absorb 50% of the impact force,
Hence each rail will absorb 25% of the Favg.

P_M = 0.25 * Favg

P_M = 0.25 * 300.30x103 N

$P_M = 75.075 \times 10^3$ N

Empirical relationship for predicting forces during crush is given by,

$P_M = 386\ t^{1.86}\ b^{0.14}\ \sigma_y^{0.57}$

Were,

P_M = Static mean crush force (N)

t = Material thickness (mm)

b = Section width and height (mm)

σ_y = Material yield stress (N/mm^2)

Substituting this value in the empirical formula

Also, $\frac{b}{t}$ = 65; b = 65t

$P_M = 386\ t^{1.86}\ (65t)^{0.14}\ \sigma_y^{0.57}$

$\therefore t^2 = \frac{75.075*10^3}{386*65^{0.14}*220^{0.57}}$

t = 2.238mm

b = 65t → b = 145.503mm

Mass of a single rail = cross sectional area of rail x $length_{steel}$ x $Density_{steel}$

Length of Front Mid-Rail = 1050mm

Mass of 2 rails = 2 x (4 x 2.238 x 145.503) x 1050 x 7.89 $x10^{-6}$

Mass of both rails = 21.581 kg

2. For Octagonal Cross-section

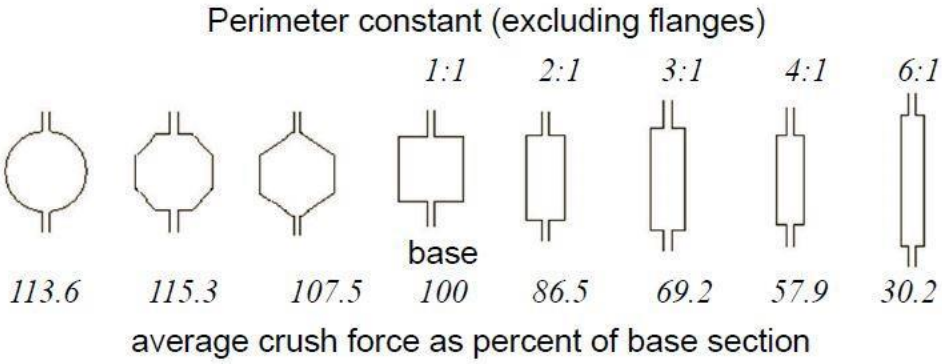


Figure 39 Section Shape effect on Average Crush Force

Now for octagonal cross-section,

The average crush force required is 1.153 times the average force required to crush square cross-section.

$$P'_M = \frac{75075}{1.153}$$

$$P'_M = 65.112x10^3 N$$

Now substituting this value of P'_M in the empirical formula.

$P_M = 386\ t^{1.86}\ b^{0.14}\ \sigma_y^{0.57}$

Here, $P_M = P'_M$ [b = 65t]

$P'_M = 386 *(65t)^{1.86} *b^{0.14} *\sigma_y^{0.57}$

$\therefore t^2 = \frac{65.112*10^3}{386*65^{0.14}*220^{0.57}}$

t = 2.084mm ; b = 65t

b = 135.505mm

Mass of a single rail = cross sectional area of rail x $length_{rail}$ x $Density_{rail}$

Length of rail = 1050 mm

Mass of 2 rails = 2 x (4 x 2.084 x 135.505) x 1050 x 7.89 $x10^{-6}$

Mass of both Rails = 18.715kg

(b) Design of the Front Upper-Rail for Energy Absorption

Assume the two vehicle upper rails will provide 25% of the required average crumpling force, FAVE. The material is the same mild steel specified in Part (a). For a width to thickness ratio, b/t = 68, determine the size of a pentagon section required to generate the required reactive force. Compute the mass of the two rails.

As mentioned in the question,

The front mid rails absorb 25% of the impact force,

Hence each rail will absorb 12.5% of the Favg.

P_M = 0.125* Favg

P_M = 0.125 * 300.30x103 N

$P_M = 37.573x10^3$ N

<u>1. For Pentagon Cross-section</u>

Now for pentagon cross-section,

The average crush force required is 1.0375 times the average force required to crush square cross-

section.

$P'_M = \frac{37573}{1.0375} = 36.173 \text{ x } 10^3$ N

Now substituting this value of P'_M in the empirical formula.

$P_M = 386\ t^{1.86}\ b^{0.14}\ \sigma_y^{0.57}$

Here, $P_M = P'_M$

& b = 68t

$P'_M = 386 *(68t)^{1.86} *b^{0.14} *\sigma_y^{0.57}$

$\therefore t^2 = \frac{36.173*10^3}{386*68^{0.14}*220^{0.57}}$

t = 1.548mm

b = 68t → b = 105.327mm

Mass of a single rail = cross sectional area of rail x $length_{rail}$ x $Density_{rail}$

Length of rail = 1210 mm

Mass of 2 rails = 2 x (4 x 1.548 x 105.327) x 1210 x 7.89 $x10^{-6}$

Mass of both Rails = 12.452 kg

Part 4B: Development of Rear Barrier Requirements

For design for a rear moving barrier, we consider the impact conditions stated above in Part 4A. Assume that two rear mid-rails provide 50% of the average force required. For a hexagonal cross-section design with b/t = 70, compute the section size needed to react the pact force. Assume the yield stress of the steel is 220 N/mm^2. Determine the mass of the two rear mid-rails.

The process to size the rear-energy-absorbing structure and reaction structure is the same as with the front barrier. Because this is a test for fuel tank integrity, we do not use cabin acceleration as the main requirement as in the front barrier. The main requirement here is to maintain the integrity of the fuel tank, so all crushes must occur behind the tank. In preliminary design, we assume the fuel tank is between the rear wheels and extends to the back of the rear wheel. The available crush space, Δ, is then the distance from the back of the rear wheel to the end of the car.

Crushable space = Δ =0.75m

Initial Velocity = V_0 = 13.88 m/s

$\sigma = 220\ N/mm^2$

Therefore, calculating the equivalent velocity after the impact:

$$V_{eqn} = V_o\sqrt{\frac{M_1}{M_1+M_2}}$$

Here,

M_1 = Mass of vehicle = 1911.03kg (given in the question)
M_2 = Mass of barrier = 1365kg (FMVSS 301)

$$V_{eqn} = V_o\sqrt{\frac{1911.03}{1911.03+1365}}$$

V_{eqn} = 10.601 m/s

$F_{avg} = \frac{M_1 * V_{eq}^2}{2 * \Delta}$

F_{avg} = 143.176KN

As mentioned in the question,

The rear mid rails provide 50% of the average force,

Hence each rail will provide 25% of the F_{avg}

P_M = 0.25 * Favg

P_M = 0.25 * 143.176x103 N

P_M = 35.793 x 10^3 N

1. For Hexagonal Cross-section

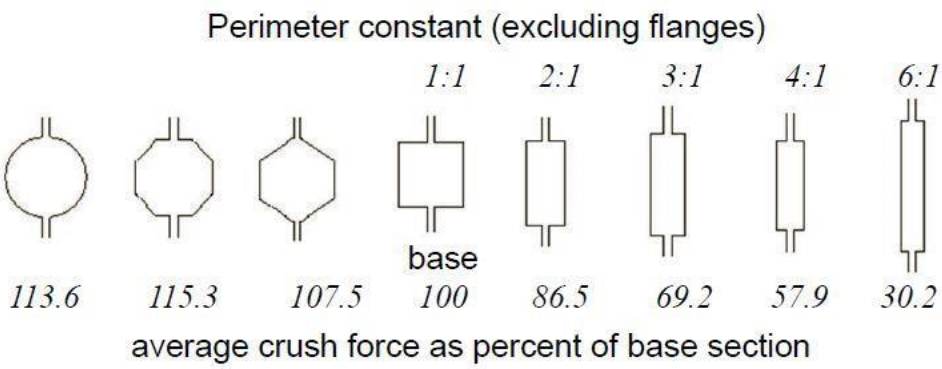


Figure 40 Section Shape effect on Average Crush Force

For hexagonal cross-section,

The average crush force required is 1.075 times the average force required to crush square cross-section.

$P'_M = \frac{35.793 * 10^3}{1.075}$

P'_M = 33.296 x 10^3 N

Now substituting this value of P'_M in the empirical formula.

$P_M = 386\ t^{1.86}\ b^{0.14}\ \sigma_y^{0.57}$

Here, $P_M = P'_M$ [b = 70t]

$P'_M = 386 * (70t)^{1.86} * b^{0.14} * \sigma_y^{0.57}$

$\therefore t^2 = \frac{33.296 * 10^3}{386 * 70^{0.14} * 220^{0.57}}$

t = 1.482 mm

b = 70t → b = 103.803 mm

Mass of a single rail = cross sectional area of rail x $length_{rail}$ x $Density_{rail}$

Length of rail = 1050mm

Mass of 2 rails = 2 x (4 x 1.548 x 103.803) x 1050 x 7.89 $x10^{-6}$

Mass of both Rails = 10.19 kg

Part 5: Design of BIW Structural Cross-Members to Prevent Structural Collapse

The objective of this part of the project is to consider the design of two BIW cross members that must be designed for maximum strength to prevent intrusion into the passenger compartment by road environment external forces. Another objective is to design the maximum load capacity of the instrument panel cross-member beam to react impact loads from a frontal collision is also considered.

Part 5A: Impact Design for the Front Roof Header

The illustration below shows the result of an impact between a vehicle traveling to the left a freeway speed with a wheel that had become detached from a truck moving in the opposite direction. The originally straight roof header is modeled as three rigid links and four plastic hinges. Assume $M_p = 20 \text{ X } 10^5$ N-mm for all the joints. The deformed geometry is shown. Note that the link between Joints 2 and 3 are assumed to remain horizontal, and all deformation occurs in-plane. Compute peak load, Fp, using limit analysis. Apply small deformation and angle assumptions.

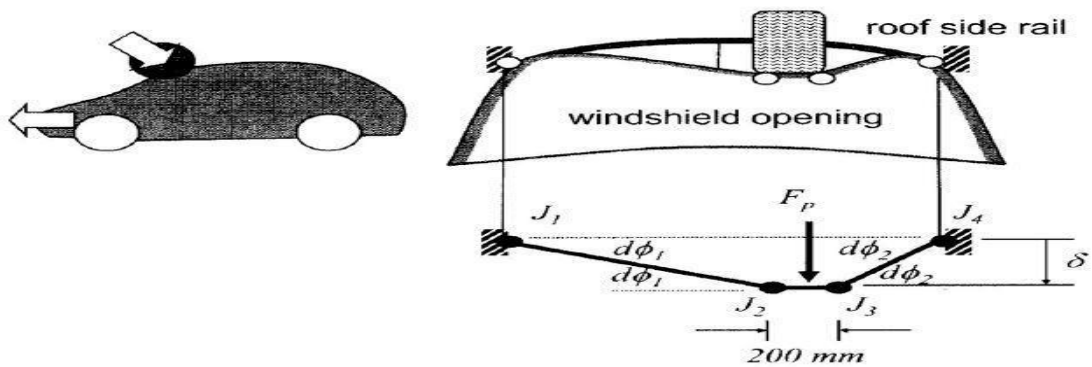


Figure 41 Front Roof Header Loading Condition

$M_P = 20x10^5$ N-mm (for all the joints)

Let the angular rotation for each hinge be dϕ

$\therefore$ F_p x δ = (M_P x dϕ1) +((M_P x dϕ1)+ (M_P x dϕ2)+ (M_P x dϕ2)

Using Trigonometric relations for small angle approximations,

$d\phi 1 = \tan^{-1}(\frac{\delta}{1025}) = (\frac{\delta}{1025})$

$d\phi 2 = \tan^{-1}(\frac{\delta}{325}) = (\frac{\delta}{325})$

Hence,

$\therefore$ F_p x δ = (M_P x dϕ1) +((M_P x dϕ1)+ (M_P x dϕ2)+ (M_P x dϕ2)

$\therefore$ F_p x $\delta = 2M_p$ x $(\frac{\delta}{1025} + \frac{\delta}{325})$

Since δ =1 mm, F_p comes out to,

$F_p = 2x20x10^5$ x $(\frac{1}{1025} + \frac{1}{325})$

F_p = 16.210 KN

Part 5B: Impact Design for the Rear Roof Header

Other types of external impact to the rear of the roof structure can be incurred by falling debris, such as tree limbs. The rear roof header can be modeled once again as three rigid links and four plastic hinges, but with a modified contact zone of 150 mm, centred on the rear header span, as shown below. Assuming $M_p = 17x10^5$ N-mm for all the joints, compute he peak load, FP, using limit analysis. Make the same assumptions as in Part 5A.

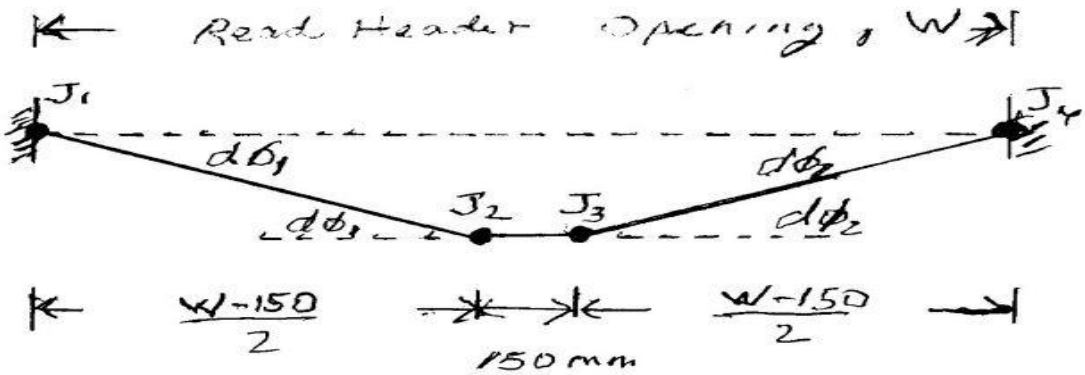


Figure 42 Rear Roof Loading Condition

Designed roof width, w = 1550mm $M_p = 17x10^5$ N-mm (for all the joints)

Let the angular rotation for each hinge be dϕ

$\therefore$ F_p x δ = (M_P x dϕ1) +((M_P x dϕ1)+ (M_P x dϕ2)+ (M_P x dϕ2)

dϕ1 = $\tan^{-1}(\frac{\delta}{\frac{w-150}{2}})$ = $(\frac{\delta}{\frac{w-150}{2}})$

dϕ2 = $\tan^{-1}(\frac{\delta}{\frac{w-150}{2}})$ = $(\frac{\delta}{\frac{w-150}{2}})$

Hence,

$\therefore$ F_p x δ = (M_P x dϕ1) +((M_P x dϕ1)+ (M_P x dϕ2)+ (M_P x dϕ2)

$\therefore$ F_p x δ = 2M_p x $(\frac{\delta}{\frac{w-150}{2}} + \frac{\delta}{\frac{w-150}{2}})$

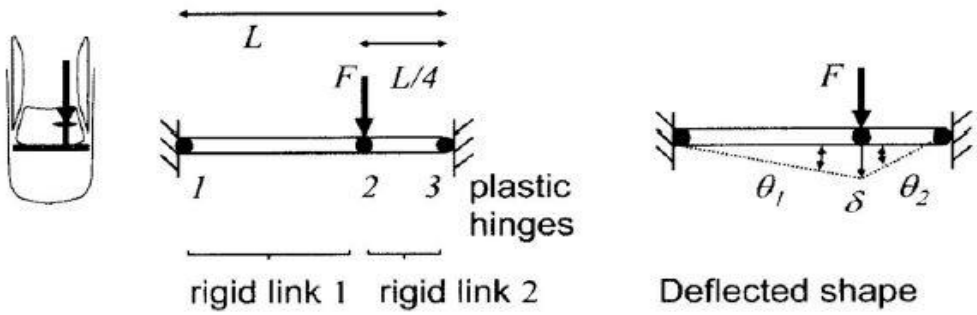
Since δ =1 mm, F_p comes out to,

F_p = 2M_p x $(\frac{1}{\frac{1550-150}{2}} + \frac{1}{\frac{1550-150}{2}})$

F_p = 9.714 KN

Part 5C: Limit Load Capacity of the Instrument Panel Beam Reacting Frontal Collision Loads

The instrument panel beam must react impact loads resulting from a frontal collision. The diver torso must be decelerated by the air bag, and inertial loads are applied in the forward direction, as shown below. Determine an expression for the peak force, F, using a limit analysis model with 3 plastic hinges and 2 links as shown. The plastic moment should be designated as MP. This moment will be determined in the following section. Apply small angles for the limit analysis.



The thin-walled section for the beam is illustrated below. For limit analysis, the effective section and stress distribution are shown. The compression face has been fully buckled and has been eliminated; the reminder of the section is at yield. Take w = 120 mm, t = 2 mm, and σ_y = 207 N/mm^2. Follow the procedure in steps (a) – (c) below to compute the plastic moment, M_p.

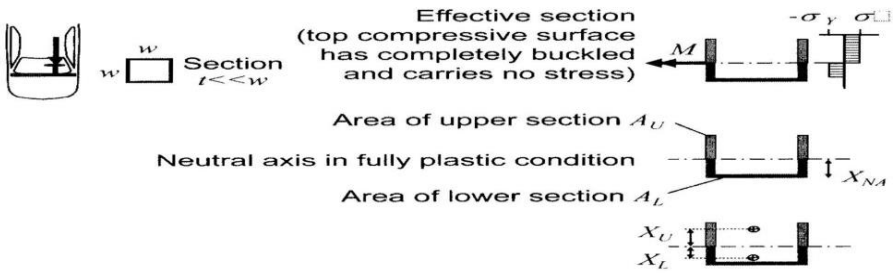
Using this value of MP, determine the limit load F derived above. Apply the instrument panel length L from your structural layout.

The thin-walled section for the beam of Exercise 6.5 is shown. For limit analysis, the effective section and the stress distribution are shown. The compression cap has been fully buckled and has been eliminated; the remainder of the section is at yield. Take $w=120\ mm$, $t=2\ mm$, also $\sigma_Y=207\ N/mm^2$.

a) Find the neutral axis position, X_{NA}, of the effective section so that $A_U = A_L$. (use $t<<w$ assumptions).

b) Find the center of gravity position of the upper area, X_U, and lower area X_L.

c) Find the plastic moment: $M_p = (\sigma_Y(A_U X_U + A_L X_L)$.



Design parameters:

w = 120 mm t = 2 mm σ_y = 207N/mm^2

Using Trigonometric relations for small angle approximations,

$d\phi 1 = \tan^{-1}(\frac{\delta}{1162.5}) = (\frac{\delta}{1162.5})$

$d\phi 2 = \tan^{-1}(\frac{\delta}{387.5}) = (\frac{\delta}{387.5})$

Considering $A_U = A_L$

$2t\,(w - X_{NA}) = 2t\,X_{NA} + t(w-2t)$

$X_{NA} = \frac{w-2t}{4} = 31mm$

$X_{NA} = 31mm$

$A_U = 2t\,(w - X_{NA})$

$A_U = 356mm^2$

$X_U = \frac{w - X_{NA}}{2}$

$X_U = 44.5mm$

$X_L = \frac{X_{NA}}{2}(2t * X_{NA}) + \left(X_{NA} - \frac{t}{2}\right)(w - 2t) * t$

$X_L = 24.95mm$

Now we calculate M_p

$M_p = \sigma_y(A_U * X_U) + (A_L * X_L)$

$M_p = 5.118$ KN-mm

Therefore, the plastic moment = $M_p = 5.118$ KN-mm

Part 6: Total Vehicle Mass Compounding Estimate Incorporating Specific Vehicle Sub-System Mass Saving Proposals & Component Mass Increases

Three mass saving opportunities have been proposed by development teams working on the vehicle design. They are:
• Powertrain: -40 kg
• Tires & Wheels: -10 kg
• Front Control Arm: -5.75 kg per arm

During the course of the vehicle development, the following component systems had a mass increase:

• Body Structure: +32 kg
• Bumpers: +6 kg
• Closures: +10 kg

Apply the ratio method of mass compounding to compute subsystem influence coefficients, based on the mass distribution table provided in the problem description. Use these coefficients with the mass distribution values to determine the updated vehicle curb mass, after resizing for the six subsystems mass changes listed above.

Body Non-Structure	292.89 kg
Body Structure	325.91 kg
Front Suspension	70.35 kg
Rear Suspension	63.17 kg
Braking	45.94 kg
Powertrain	265.91 kg
Fuel & Exhaust	57.43 kg
Steering	20.10 kg
Tires & Wheels	93.32 kg
Electrical	66.04 kg
Cooling	38.77 kg
Bumpers	31.59 kg
Closures	64.61 kg
Vehicle Curb Mass	1436.03kg
Passengers	350 kg
Cargo	125 kg
Gross Vehicle Mass	1911.03 kg

A means to quantify the secondary mass change is using a mass-compounding model. In this model, each subsystem (denoted by i) is assigned a mass influence coefficient, $\boldsymbol{\gamma}_i$. The influence coefficient is the change in the subsystem mass when gross vehicle mass undergoes a unit change. The physical interpretation of the influence coefficient is each subsystem is sized to some degree by the mass of the vehicle, and as the vehicle mass changes the subsystem may also be resized.

a. Three mass saving opportunities have been proposed by development teams working on the vehicle design. They are:

• Powertrain: -40 kg

• Tires & Wheels: -10 kg

• Front Control Arm: -5.75 kg per arm = 11.5kg for both

Therefore, the total mass saved is 61.5kg.

Mass of the vehicle excluding the mass of the passenger & cargo is Curb Mass (M_{curb})

M_{curb} = 1436.03kg.

This resizing in response to a primary change is captured by a mass-compounding model:

$$M_v^c = M_v + \Delta + \Delta S_v$$

Where,

$S_v = \left[\frac{\gamma_i}{1-\gamma_i}\right]$

M_v = the initial vehicle mass for which the subsystems are sized.

Δ = the initial (primary) mass change for the vehicle. This is the sum of the primary changes of each subsystem, $\Delta = \Sigma\Delta_i$

Δ_i = primary change in subsystem i

M_v^c = the resized (compounded) vehicle mass

ΔS_v = the additional (secondary) mass change due to resizing subsystems

γ_i = the mass influence coefficient for subsystem i

γ_v = the mass influence coefficient for the vehicle given by

Components	Mass	γ_i	γ_i
Body Non Structure	292.89	0.15326	0
Body Structure	325.91	0.17054	0.17054
Front Suspension	70.35	0.03681	0.03681
Rear Suspension	63.17	0.03306	0.03306
Braking	45.94	0.02404	0.02404
Powertrain	265.91	0.13914	0.13914
Fuel & Exhaust	57.43	0.03005	0.03005
Steering	20.1	0.01052	0.01052
Tyres & Wheels	93.32	0.04883	0.04883
Electrical	66.04	0.03456	0
Cooling	38.77	0.02029	0
Bumpers	31.59	0.01653	0.01653
Closures	64.61	0.03381	0
	$\gamma_V =$	0.5095	0.50953
Passenger	350	0.18315	0.18315
Cargo	125	0.06541	0.06541

Body Non-Structural, Electrical, Cooling, Closures are assumed to have no dependence on Gross

Vehicle Mass.

$S_v = \left[\frac{\gamma_i}{1-\gamma_i}\right]$

S_v = 1.0387

M_v^c = 1436.03 + (-61.5) +(-61.5 x 1.0387)

M_v^c = 1310.64 kg

b. During the vehicle development, the following component systems had a mass increase:

• Body Structure: +32 kg

• Bumpers: +6 kg

• Closures: +10 kg

Therefore, the total mass increased is 48kg.

New Mass of the vehicle excluding the mass of the passenger & cargo is Curb Mass (MCURB) and

after increase in the above-mentioned sub-systems is

MCURB = 1358.64kg.

This resizing in response to a primary change is captured by a mass-compounding model:

$$M_v^c = M_v + \Delta + \Delta S_v$$

Where,

$S_v = \left[\frac{\gamma_i}{1-\gamma_i}\right]$

M_v = the initial vehicle mass for which the subsystems are sized.

Δ = the initial (primary) mass change for the vehicle. This is the sum of the primary changes of each subsystem, $\Delta = \Sigma\Delta_i$

Δ_i = primary change in subsystem i

M_v^c = the resized (compounded) vehicle mass

ΔS_v = the additional (secondary) mass change due to resizing subsystems

γ_i = the mass influence coefficient for subsystem i

γ_v = the mass influence coefficient for the vehicle given by

Components	Mass	γ_i	γi
Body Non Structure	292.89	0.16403	0
Body Structure	325.91	0.18252	0.18252
Front Suspension	58.85	0.03296	0.03296
Rear Suspension	63.17	0.03538	0.03538
Braking	45.94	0.02573	0.02573
Powertrain	225.91	0.12651	0.12651
Fuel & Exhaust	57.43	0.03216	0.03216
Steering	20.1	0.01126	0.01126
Tyres & Wheels	83.32	0.04666	0.04666
Electrical	66.04	0.03698	0
Cooling	38.77	0.02171	0
Bumpers	31.59	0.01769	0.01769
Closures	64.61	0.03618	0
	γ_v =	0.5109	0.51086
Passenger	350	0.19601	0.19601
Cargo	125	0.07	0.07

$S_v = \left[\frac{\gamma_i}{1-\gamma_i}\right]$:

S_v = 1.044

M_v^c = 1310.64 + (48) +(48 x 1.0387)

M_v^c = 1408.752 Kg.